



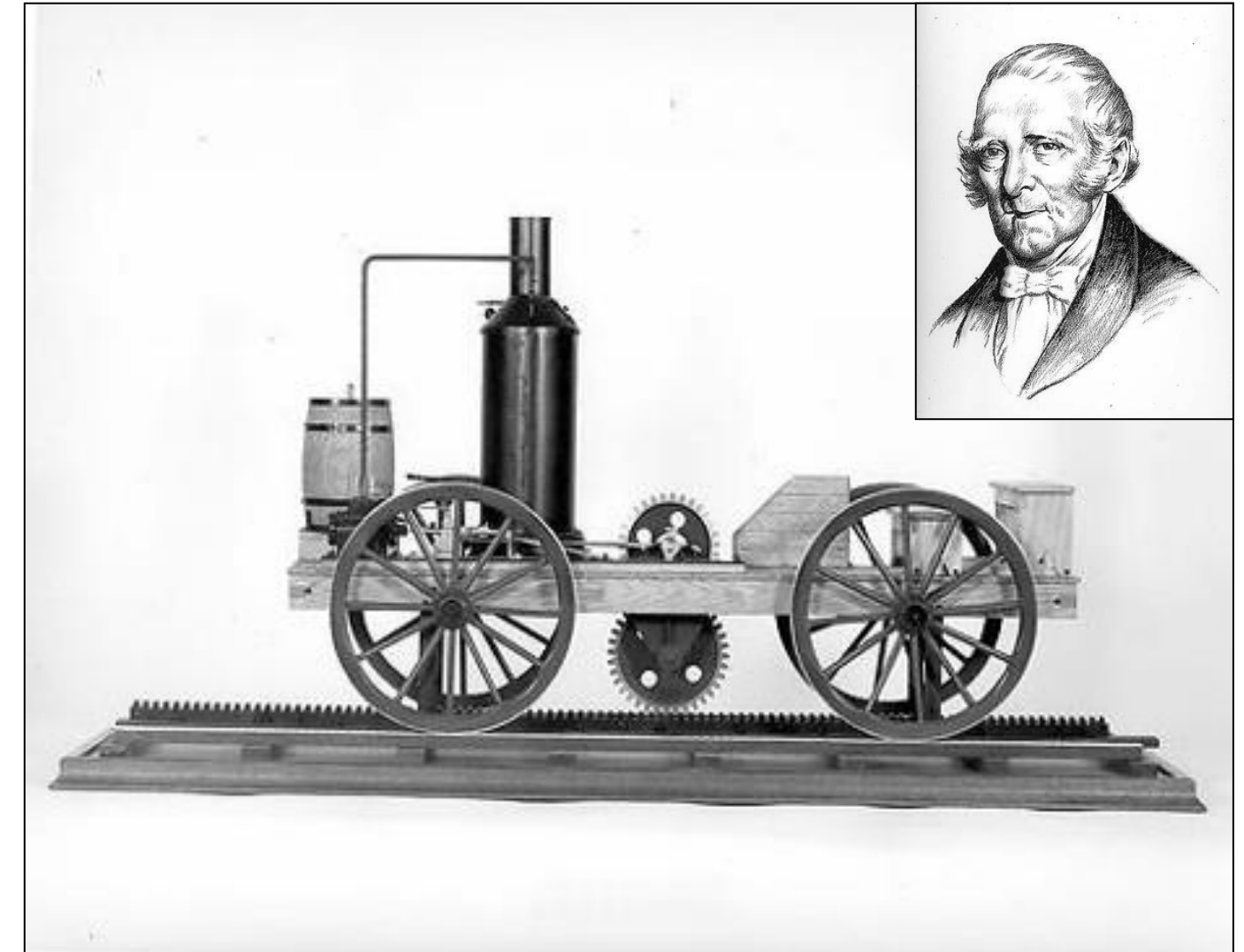
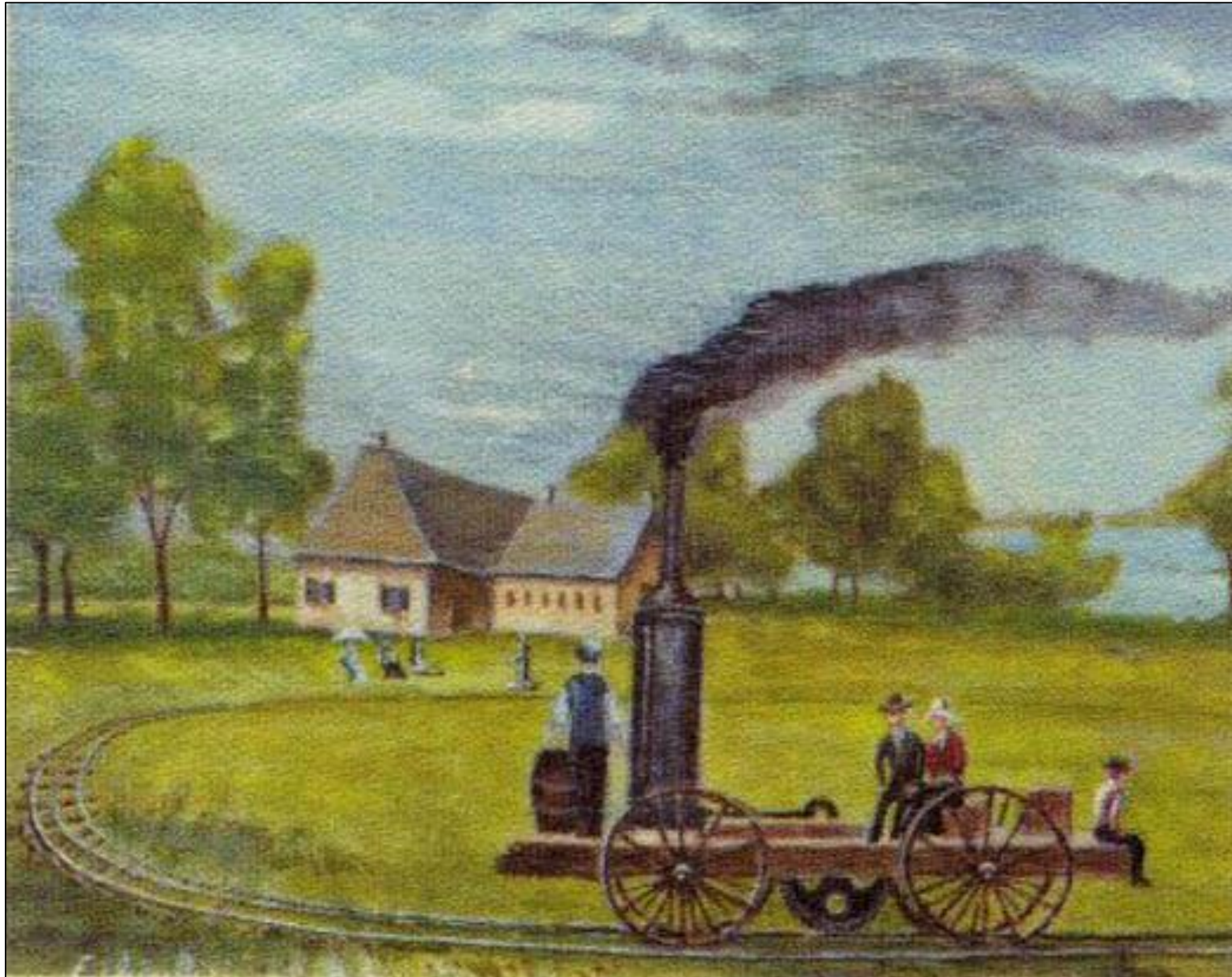
Vehicle-Track Interaction & Dynamics

Corey Pasta
Scientist, MxV Rail

DALLAS, TX



"I can see nothing to hinder a steam carriage moving on its ways with a velocity of 100 miles an hour."
Colonel John Stevens, 1812



Colonel John Steven's and a model of his locomotive

Vehicle-Track Interaction

1. A railway vehicle moving along a track is a complex system
 - Multiple degrees of freedom
 - Non-linear connections
 - Complex geometry and contact conditions at the wheel/rail interface
2. Solutions require engineering trade-offs to meet desired operating conditions



We'll approach these concepts as academics with overalls.





Overview

1. Steel Wheels on Steel Rails
2. High Speed Stability
3. Curving & Load Equalization
4. Resonance in the Vehicle-Track System
5. External Forces



(1) Steel Wheels on Steel Rails *An industry advantage with maintenance challenges*

“It still costs, on average, less than one cent for hauling a ton of freight a mile.”
1946

“Railroads are the most fuel-efficient way to move freight over land, moving one ton of freight nearly 500 miles per gallon of fuel.”

Association of American Railroads
2026



Speed & Resistance by Transport Mode

Rail: Speed + Capacity + Low Resistance

Resistance (lbs/ton)

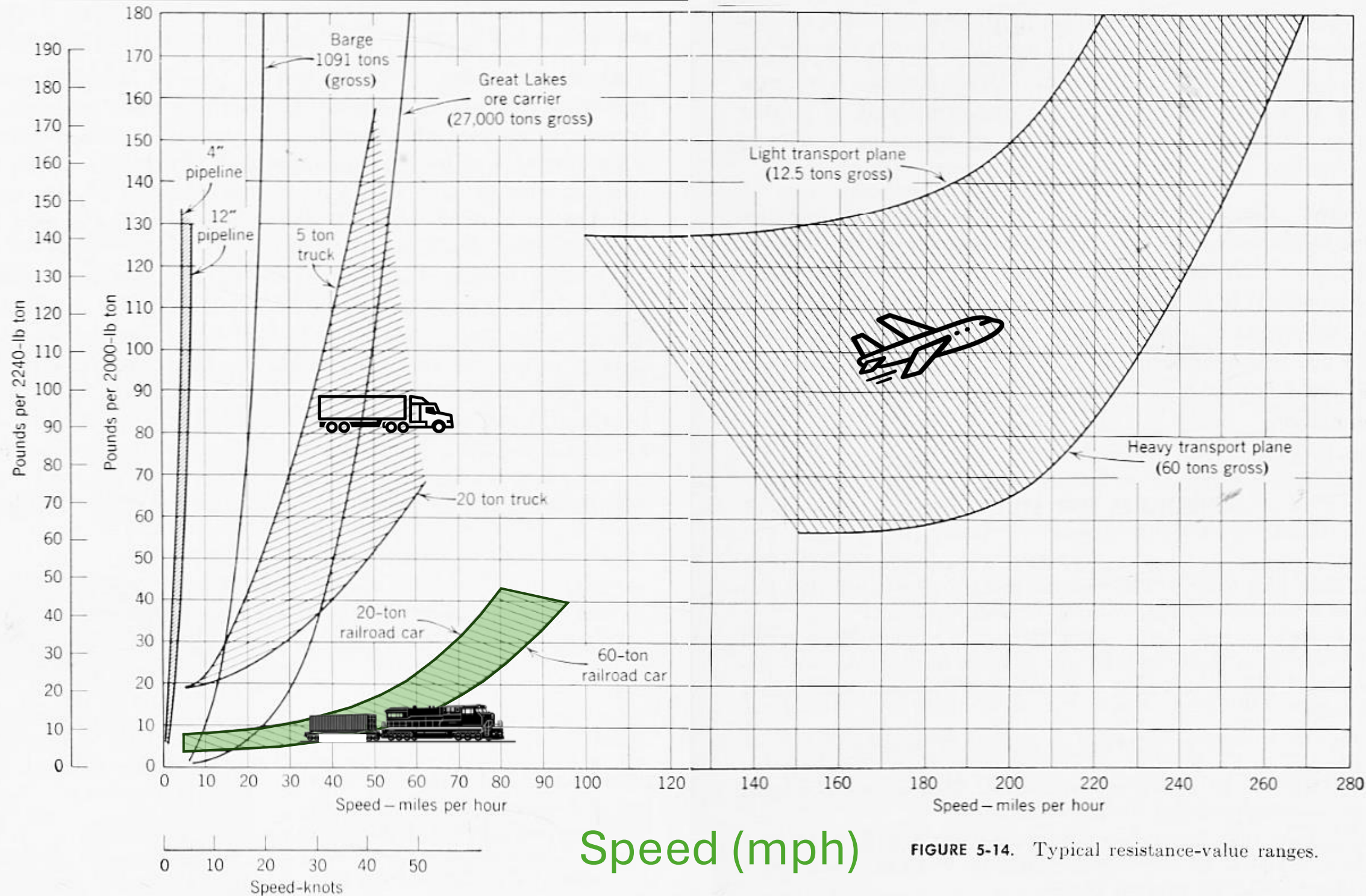


FIGURE 5-14. Typical resistance-value ranges.

Rolling resistance: μ_r



Steel wheel on rail:

$$\mu_r = 0.001$$

Truck tire on pavement :

$$\mu_r = 0.006 \text{ to } 0.01$$

Rail has a unique combination of low resistance at high capacity and speed.

An Introduction to Transportation Engineering, Hay 1961



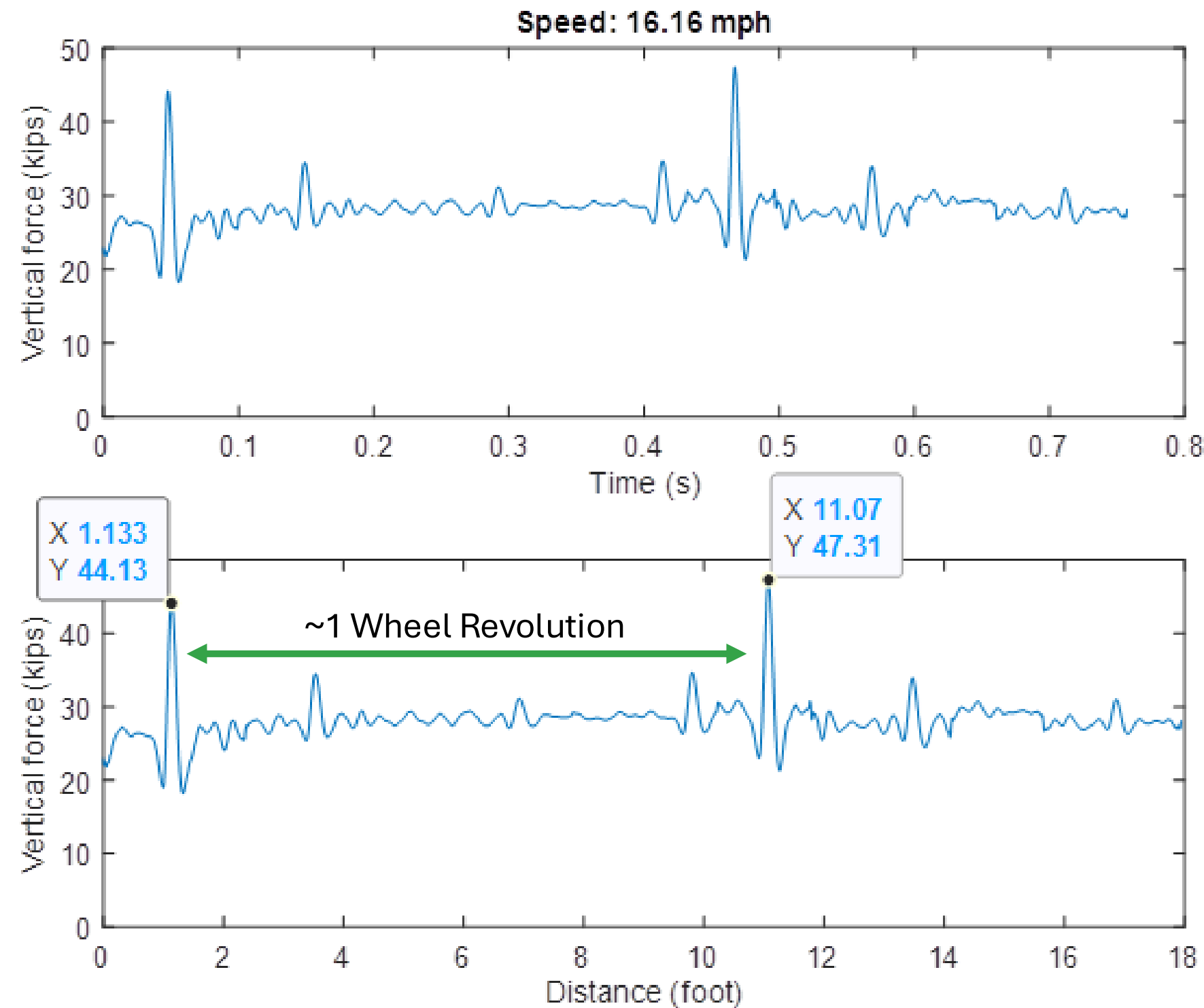
Challenges with Steel Wheels on Steel Rails

Unique Damage Modes



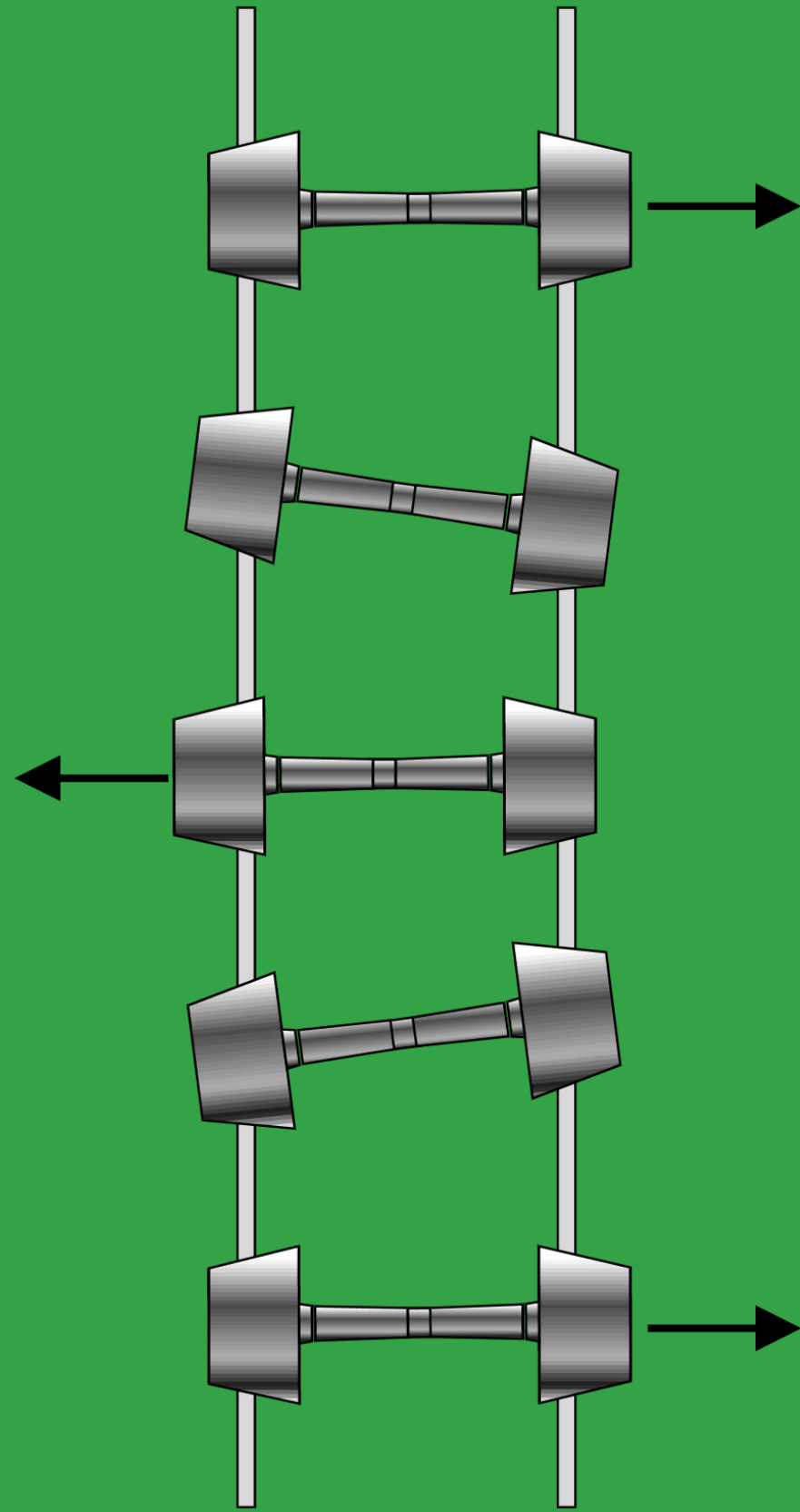


Impacts at Wheel/Rail Interface



Sample Wheel Impact Measurement

- An out-of-round wheel impacts the track with each rotation.
- Discontinuities in the rail also produce repeated impacts.
- Impacts affect both track and wheel life.



(2) High Speed Stability

Everyone Can Understand Hunting & Curving!



Photo courtesy of traveling MxV Rail Engineer



Photo courtesy RailTEC UIUC



Truck Hunting at Higher Speed



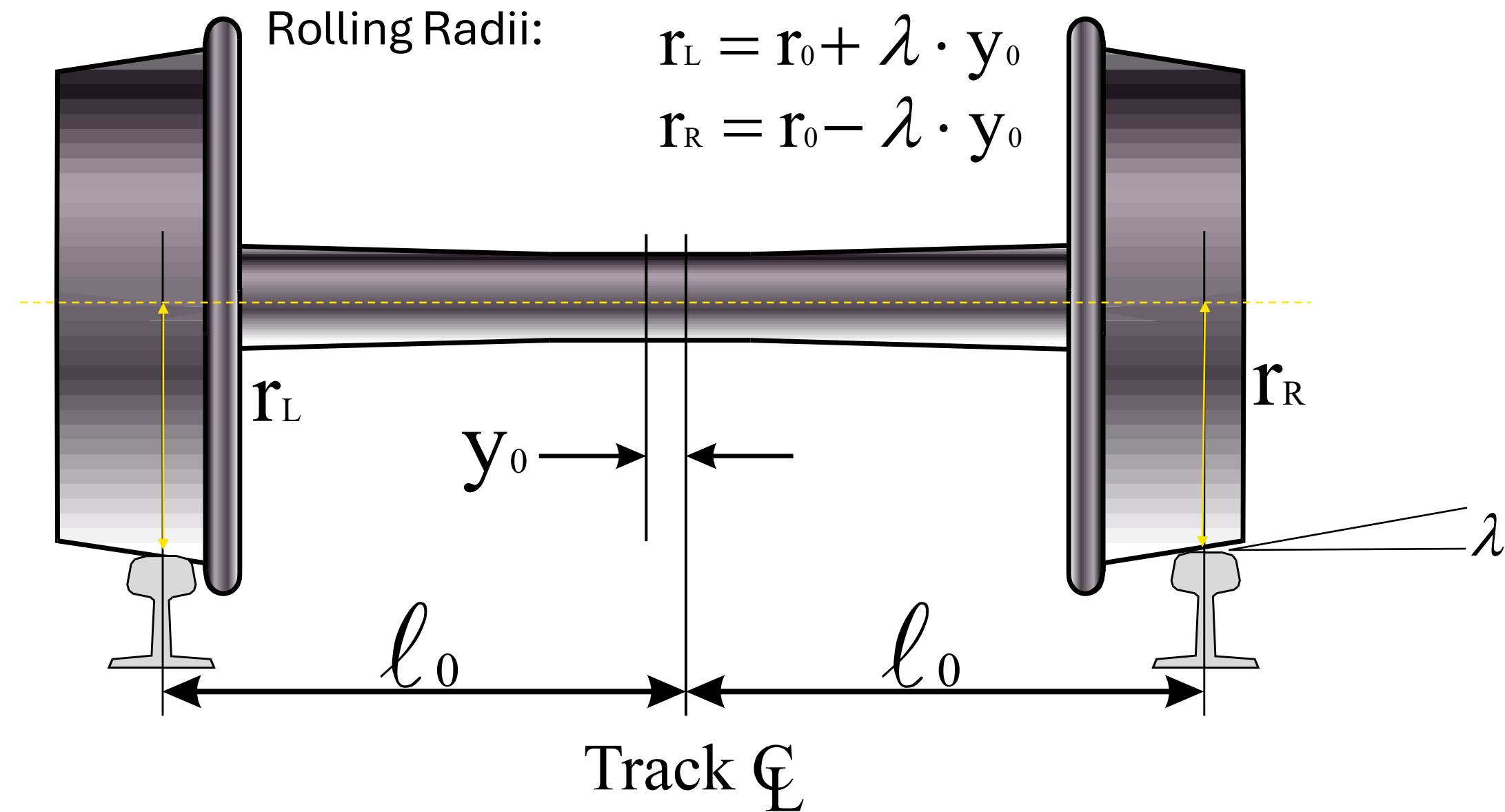
Example 1: Hopper Car



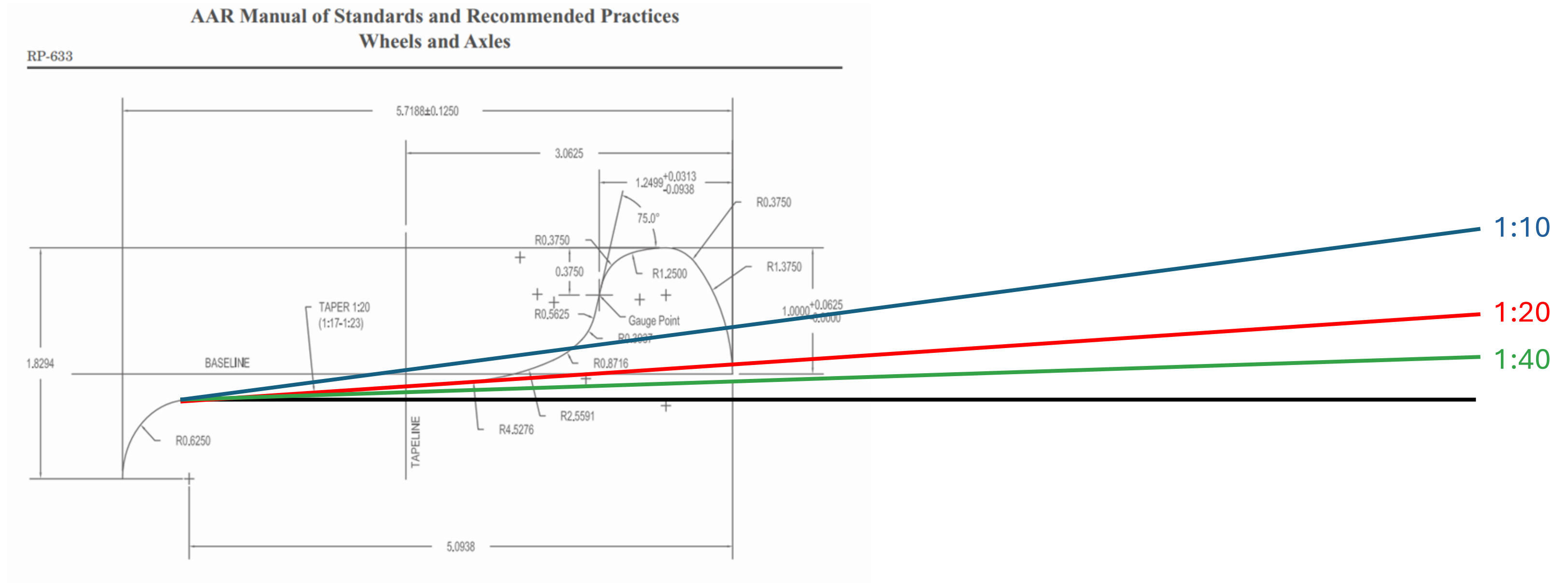
Example 2: Flat Car



Wheelset Rolling Radii



Wheelset Taper / Conicity



Kinematic Frequency and Wavelength

Derived by Klingel in 1883:

$$Wavelength = \frac{V}{f} = 2\pi \sqrt{\frac{l_o \cdot r_o}{\lambda}}$$



A free wheelset will oscillate at a wavelength inversely proportional to the square of the conicity.

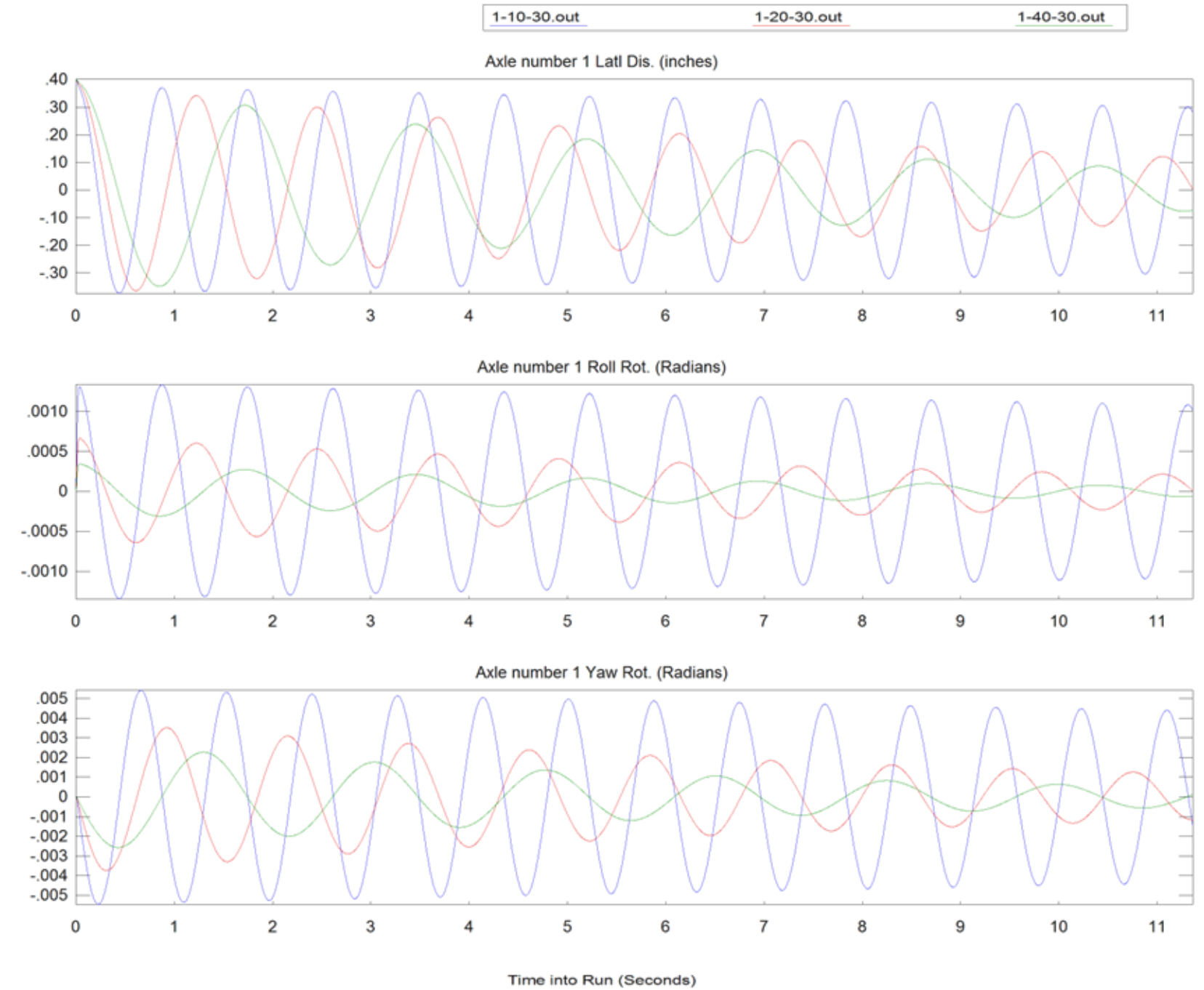
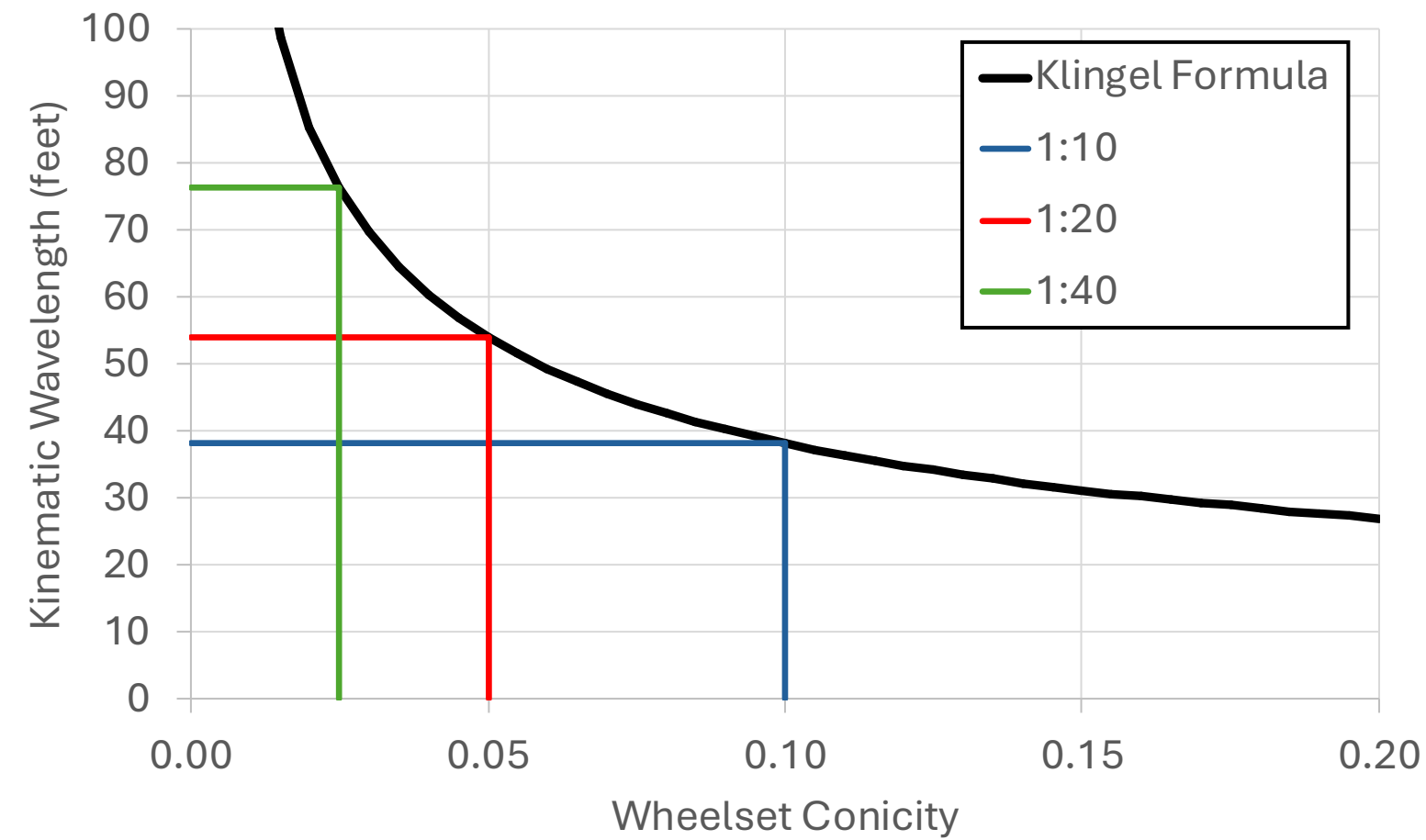
More conicity (more taper) yields a shorter wavelength (more hunting).





Free Wheelset Kinematic Wavelength

1:10 / 1:20 / 1:40 Taper at 30mph



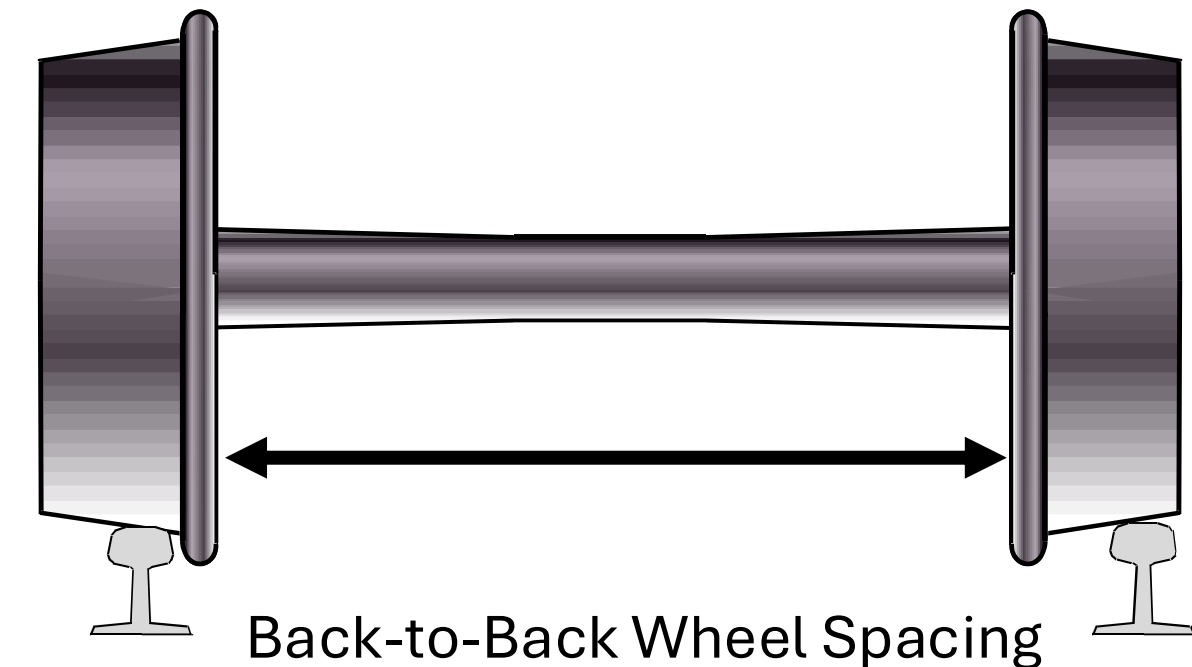


Influence of Track on Truck Hunting

Equivalent conicity increases significantly at the wheel flange root

Wheel flange root contact increases with:

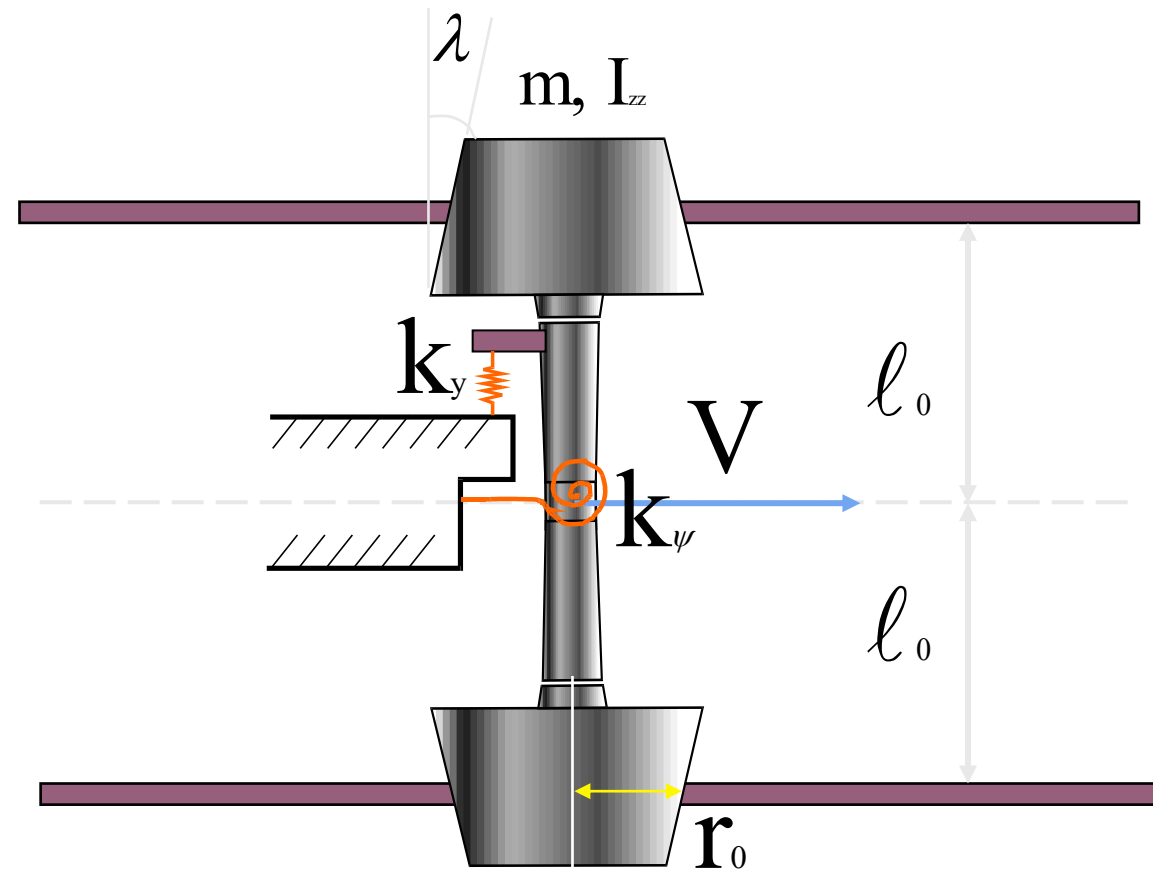
- Narrowing of effective track gage such as metal flow over the gage-side corner
- Larger back-to-back wheel spacing





How do we fix the stability problem?

Restrain motion and maintain profiles



Hunting Speed $V_C = \sqrt{\frac{r_0 l_0 (k_\psi + l_0^2 k_y)}{\lambda (I_{zz} + l_0^2 m)}}$



To increase high speed stability:

↑ k_ψ, k_y

Yaw stiffness and lateral stiffness

- Truck warp stiffness
- Truck turning moment

↓ λ

Wheel/Rail profiles, effective conicity

Watch for tight gage and high back-to back spacing

↑ l_0

Track gage

↓ I_{zz}, m

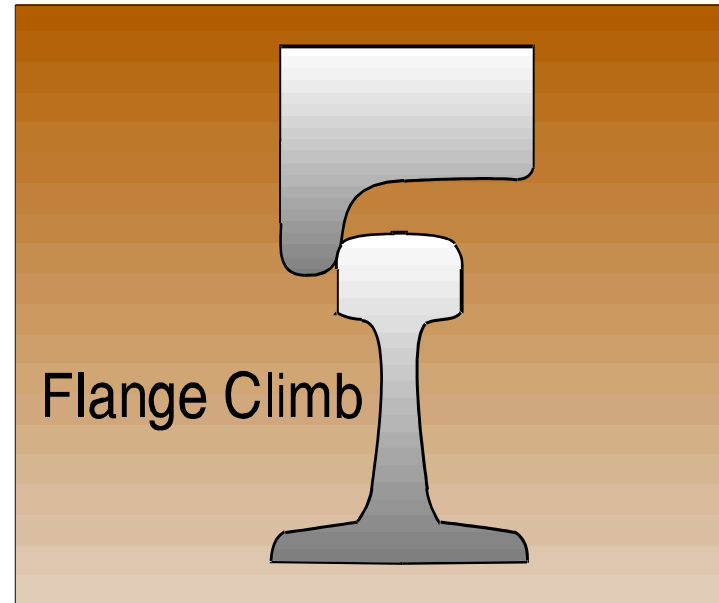
Wheelset mass and yaw inertia

↑ r_0

Wheel radius

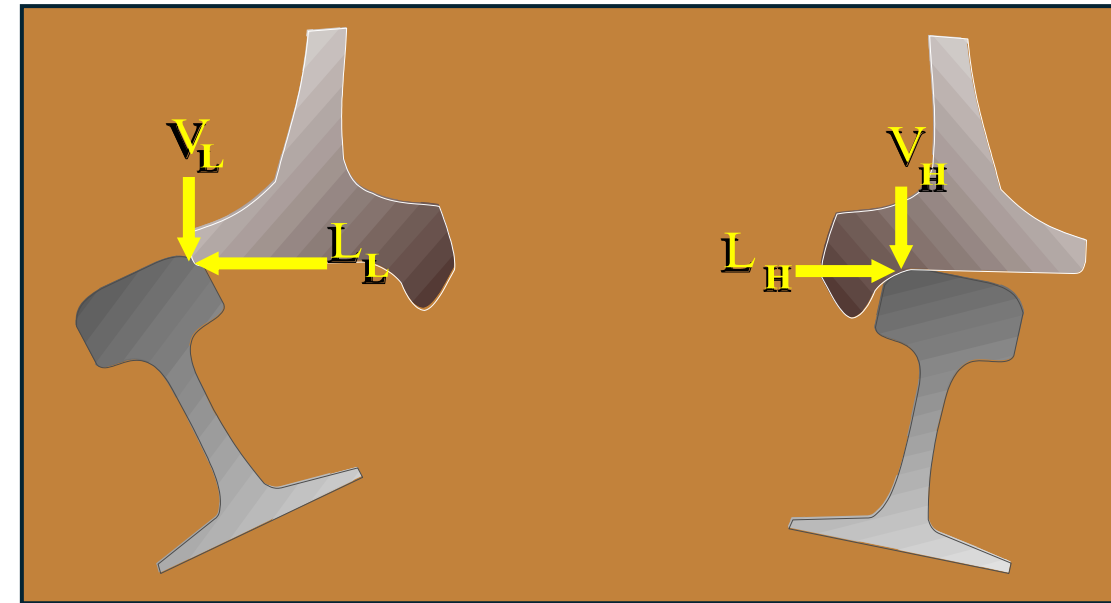


Two Common Derailment Mechanisms



Contributing Causes

- High lateral force
- Low vertical force
- Angle of attack
- Friction



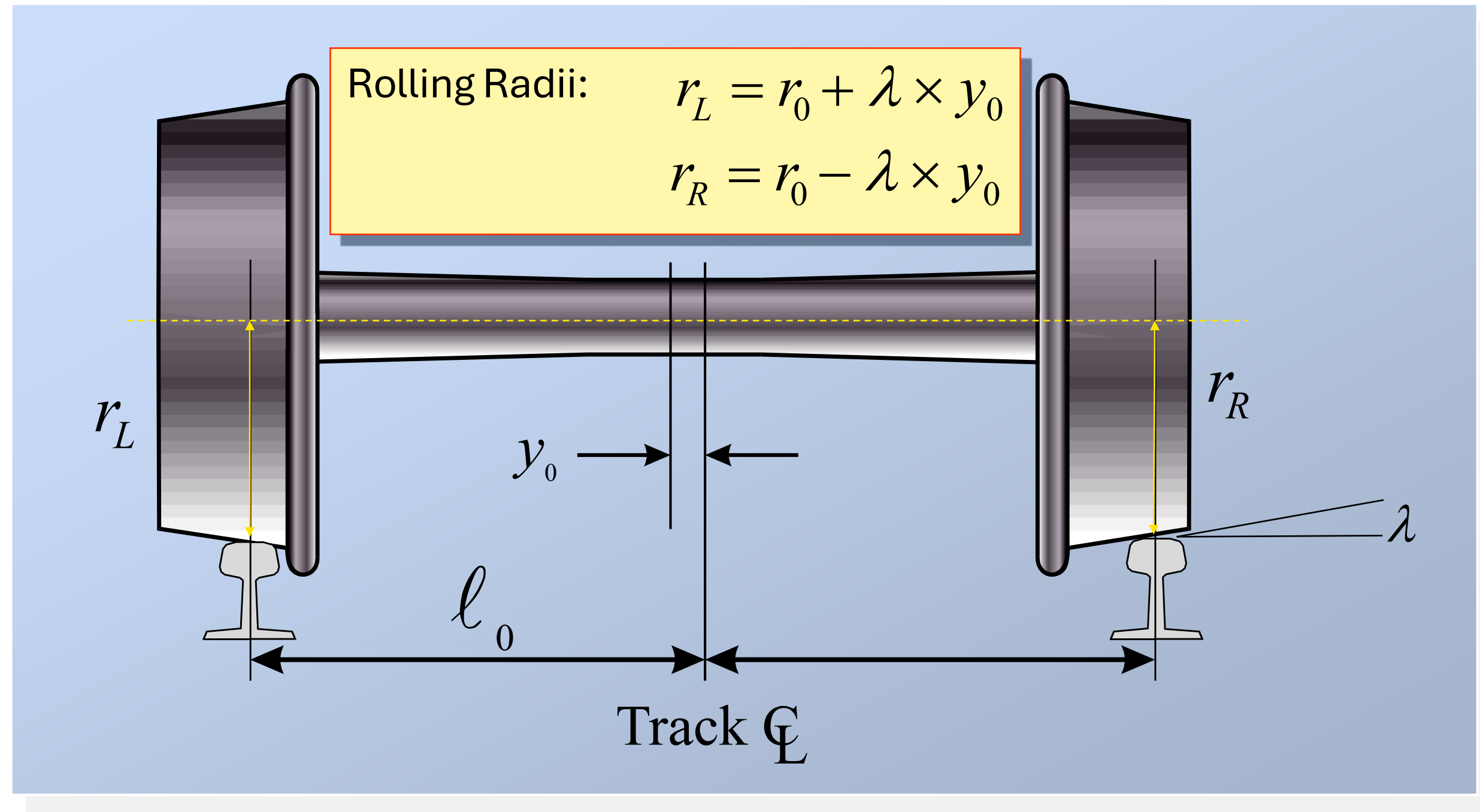
Contributing Causes

- Insufficient gage / rail restraint
- Poor truck steering (proximity of adjacent axles)
- Poor wheel/rail profile conditions
- Proximity of adjacent axles

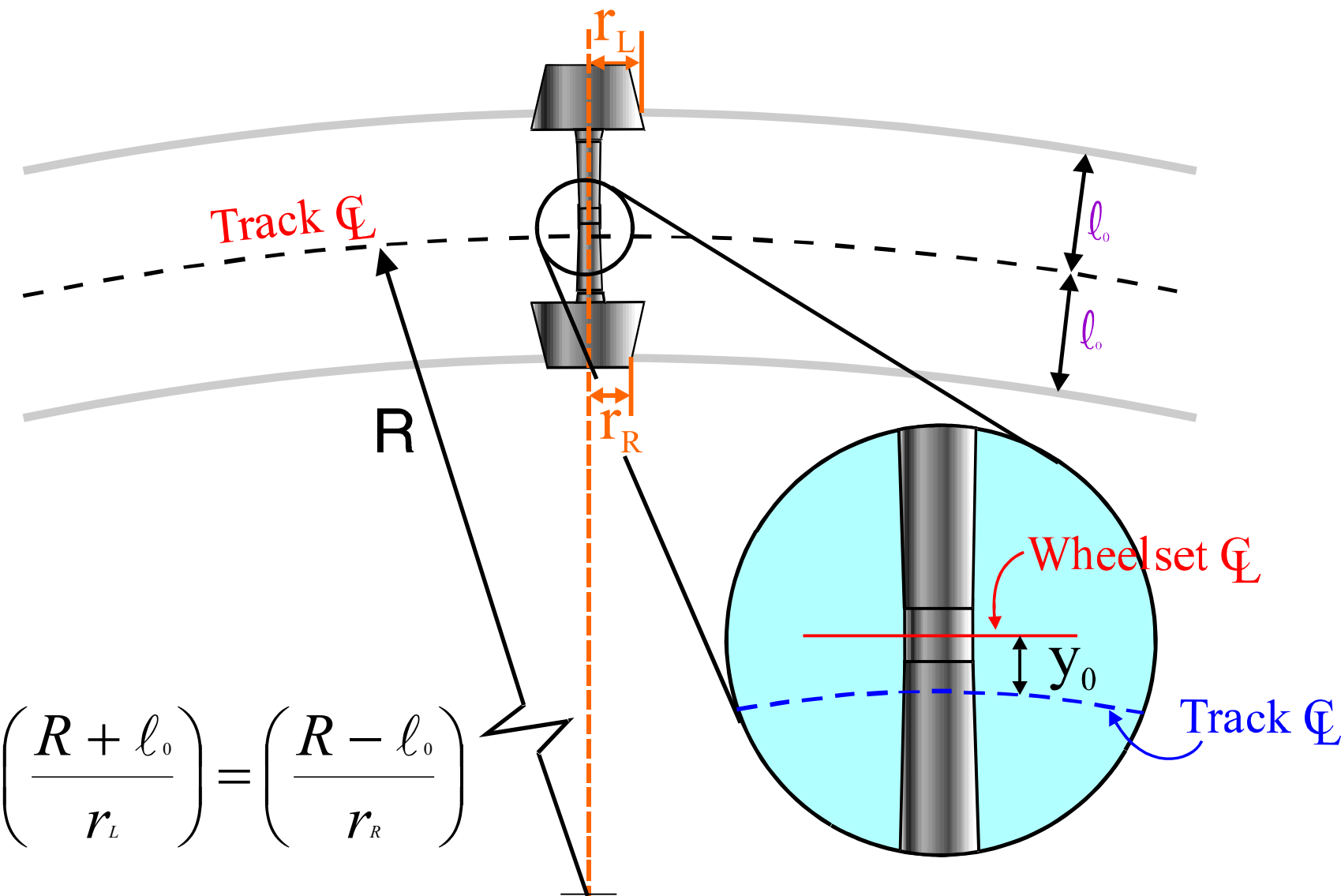


(3) Curving & Load Equalization

Wheelset Rolling Radii



Equilibrium Rolling Line



$$\left(\frac{R + \ell_0}{r_L} \right) = \left(\frac{R - \ell_0}{r_R} \right)$$

$$\left(\frac{R + \ell_0}{r_0 + \lambda \cdot y_0} \right) = \left(\frac{R - \ell_0}{r_0 - \lambda \cdot y_0} \right)$$

$$\therefore y_0 = \frac{\ell_0 r_0}{\lambda R}$$

(Redtenbacher, 1855)



Wheelsets that curve well track closer to the centerline with:

↑ λ

Wheel/rail profiles with greater conicity

↓ r

Smaller wheels

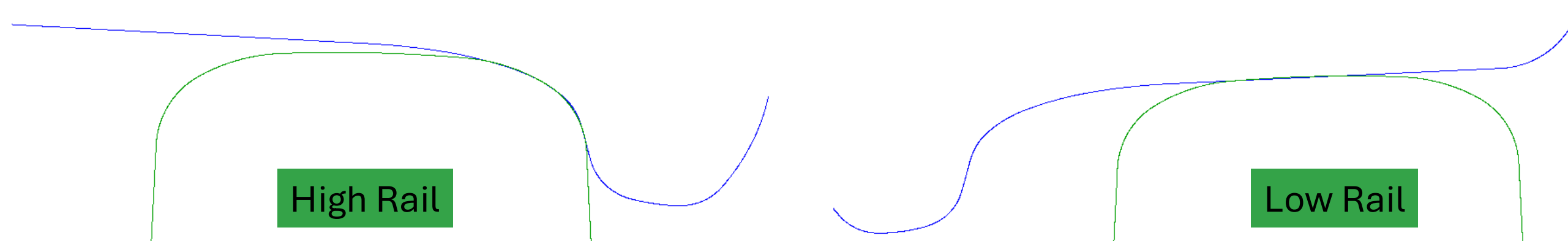
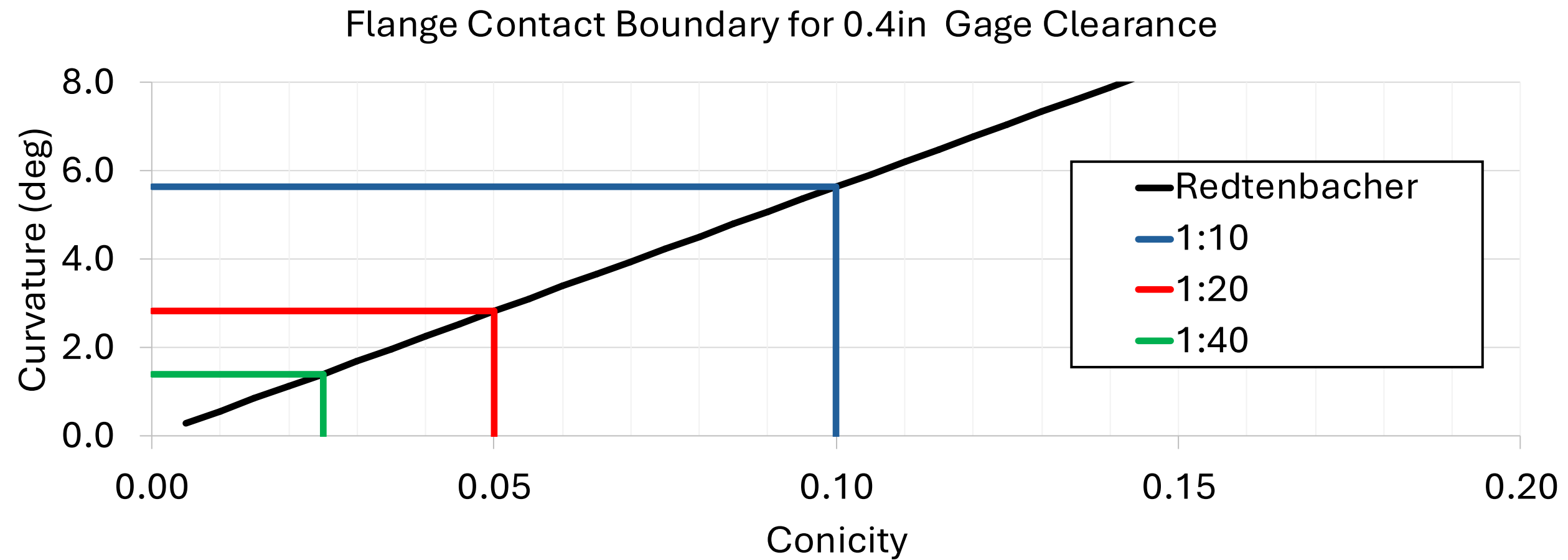
↓ ℓ_0

Narrower track gage





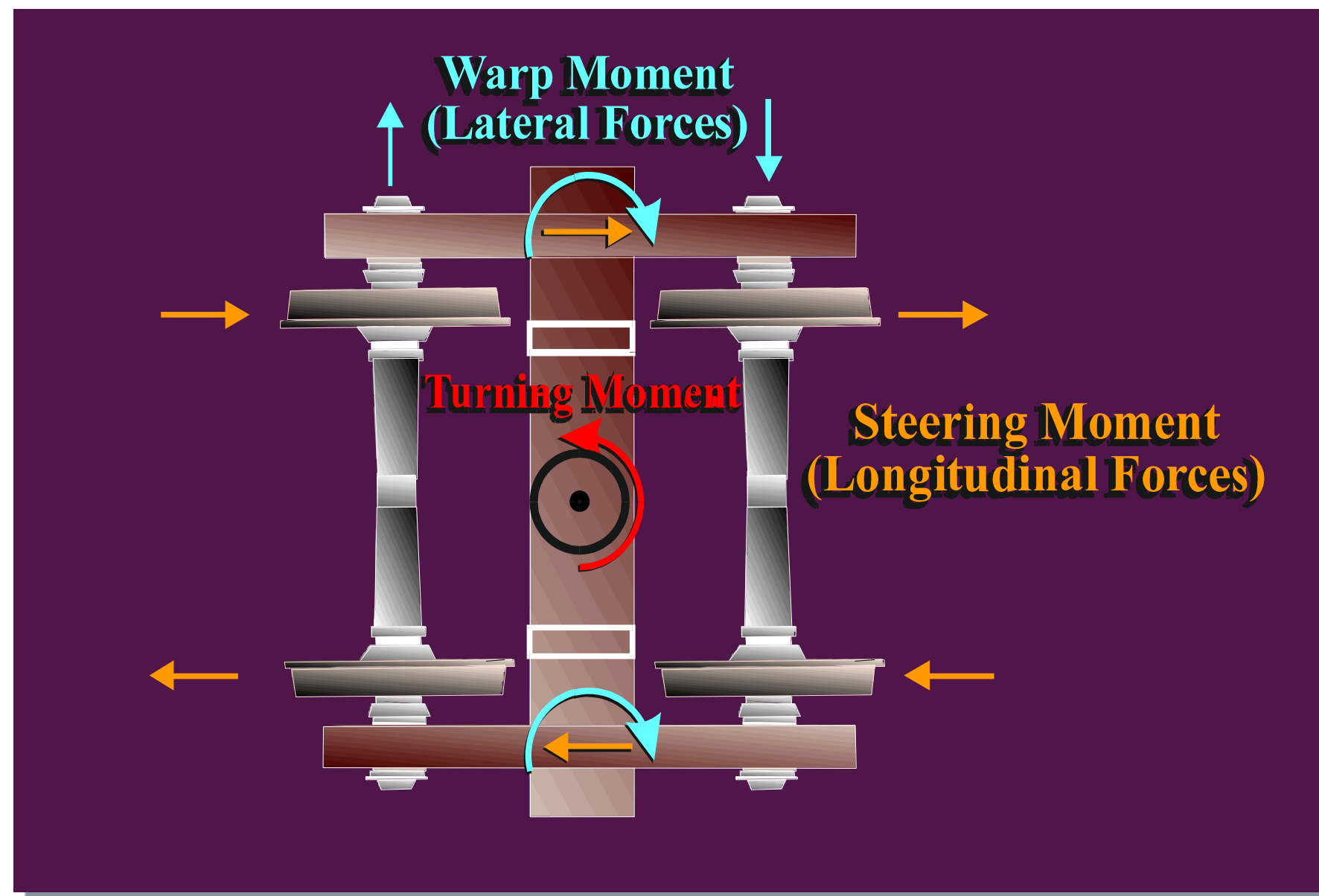
Free Wheelset Flange Contact Boundary



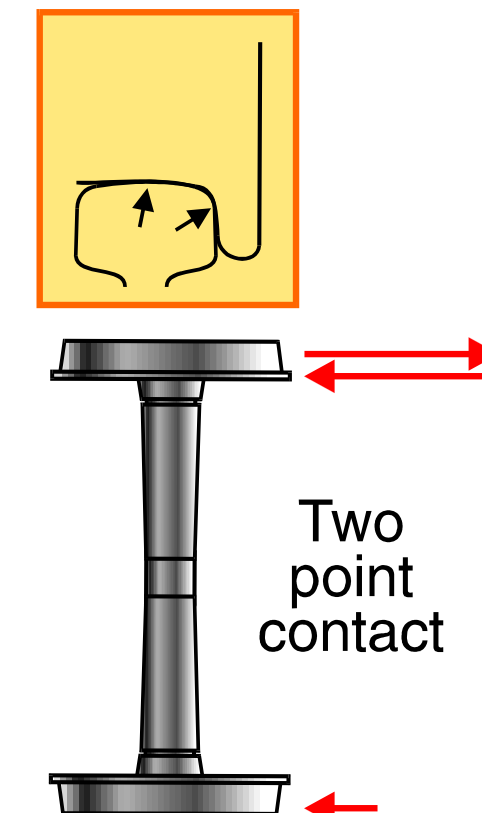
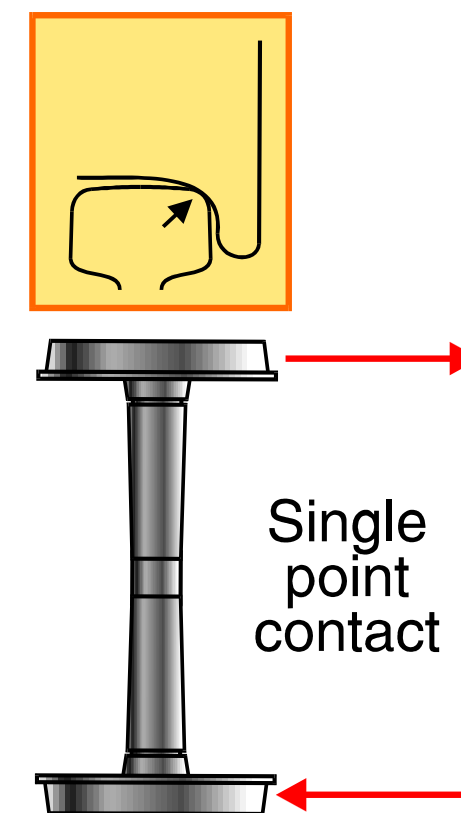
Example: Lateral Wheelset Position in 4-degree Curve



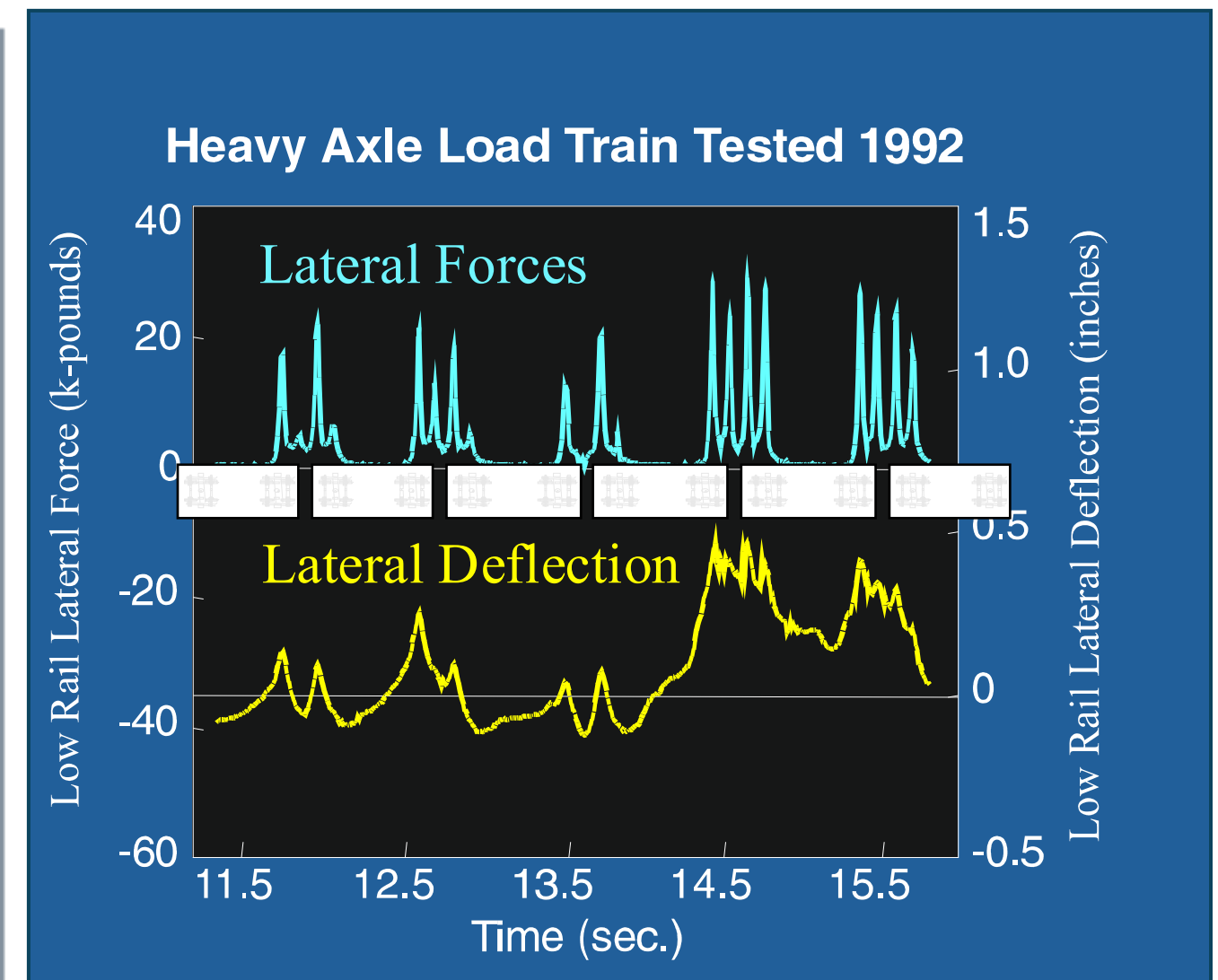
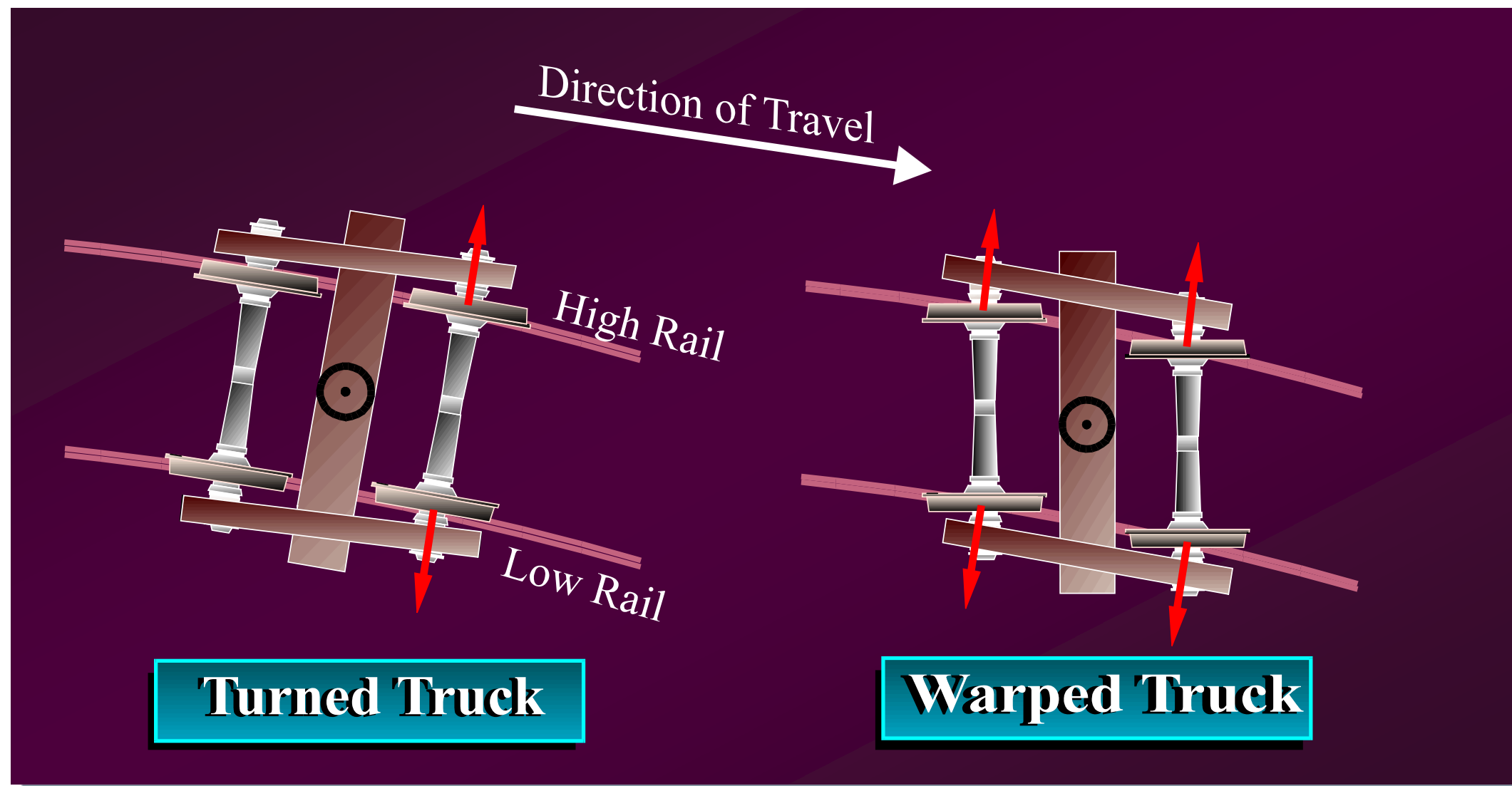
Truck Steering and Wheel/Rail Profiles



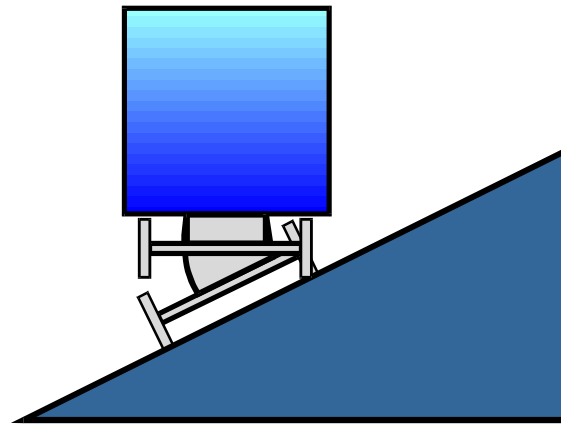
$$\text{Turning Moment} = \text{Warp Moment} + \text{Steering Moment}$$



Truck Warp and Lateral Forces

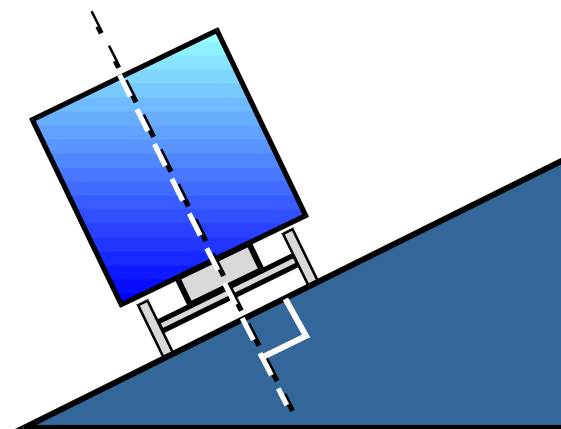


Under- and Over-Balanced Conditions



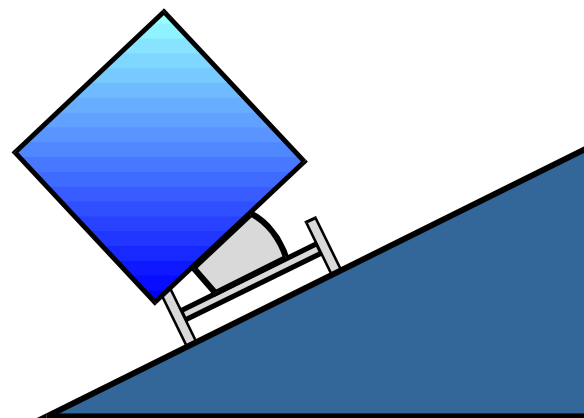
$$V > V_{Balance}$$

- Over balance
- Cant deficiency
- Increase in vertical load on high rail



$$V = V_{Balance}$$

- Balance
- Equal load on high and low rail



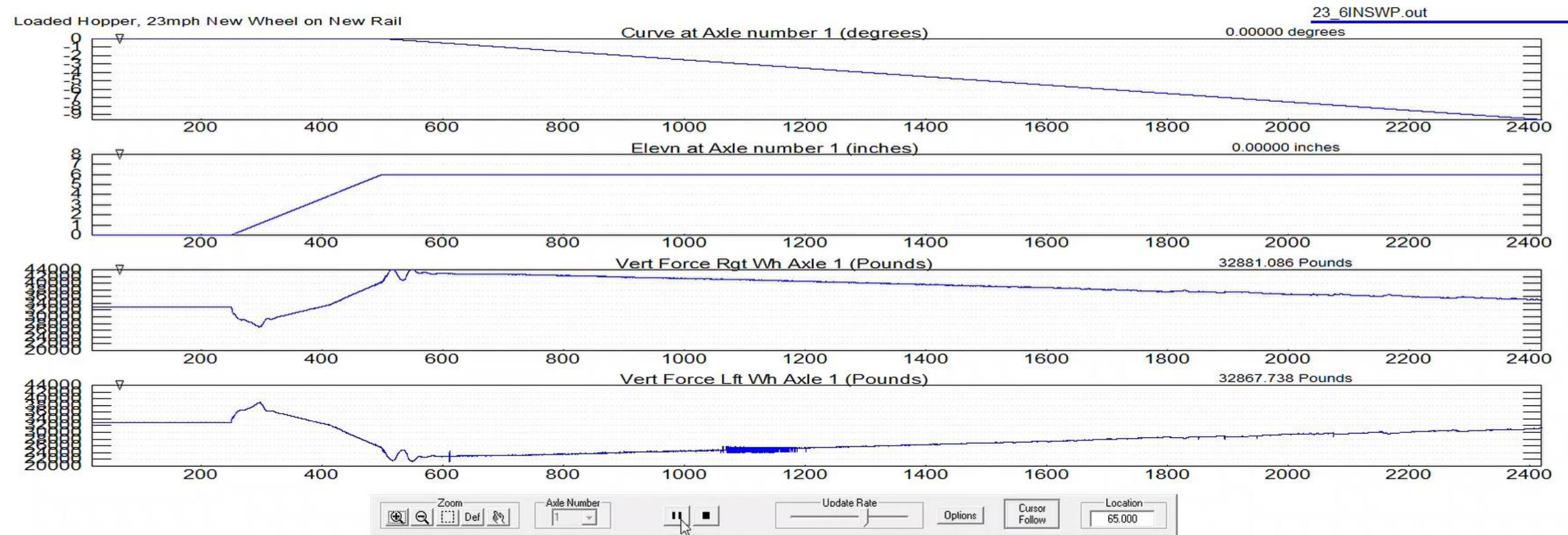
$$V < V_{Balance}$$

- Under balance
- Cant excess
- Decrease in vertical load on high rail





Truck Steering Example

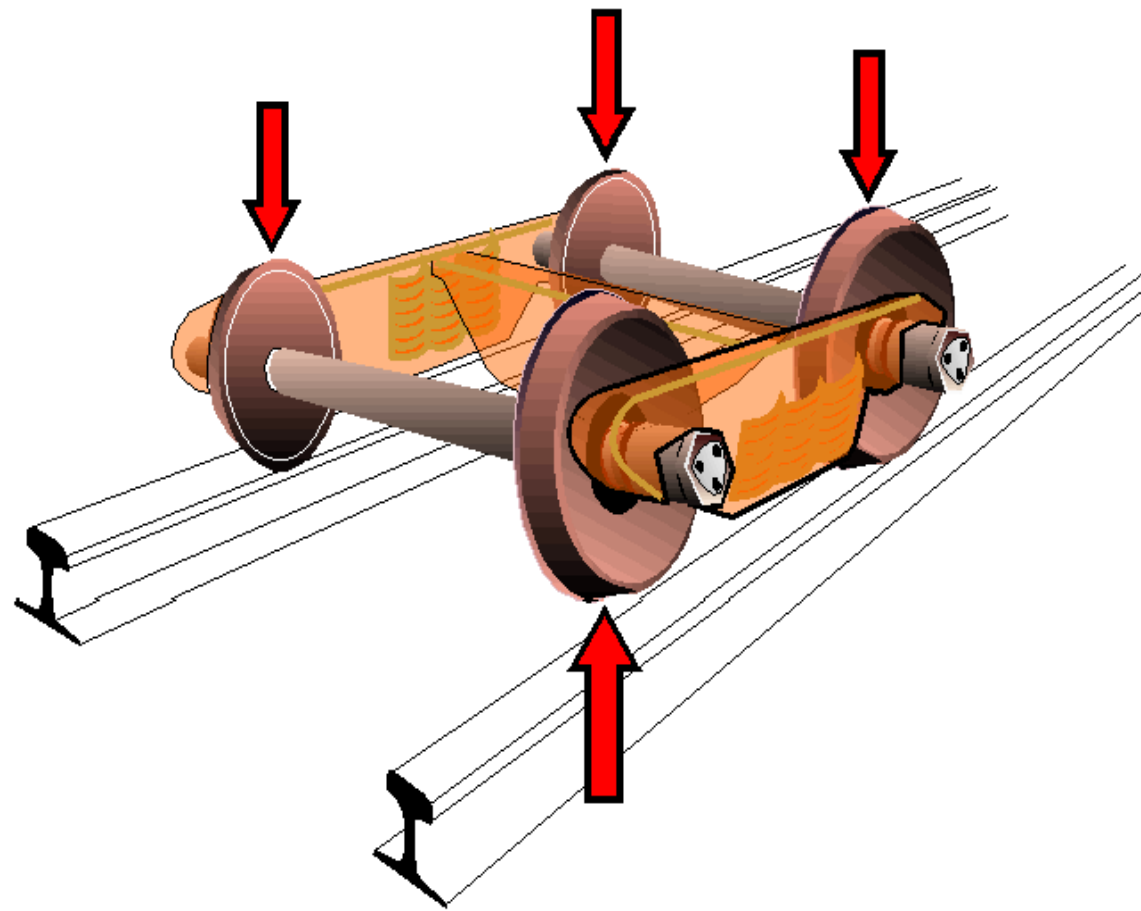


High Rail

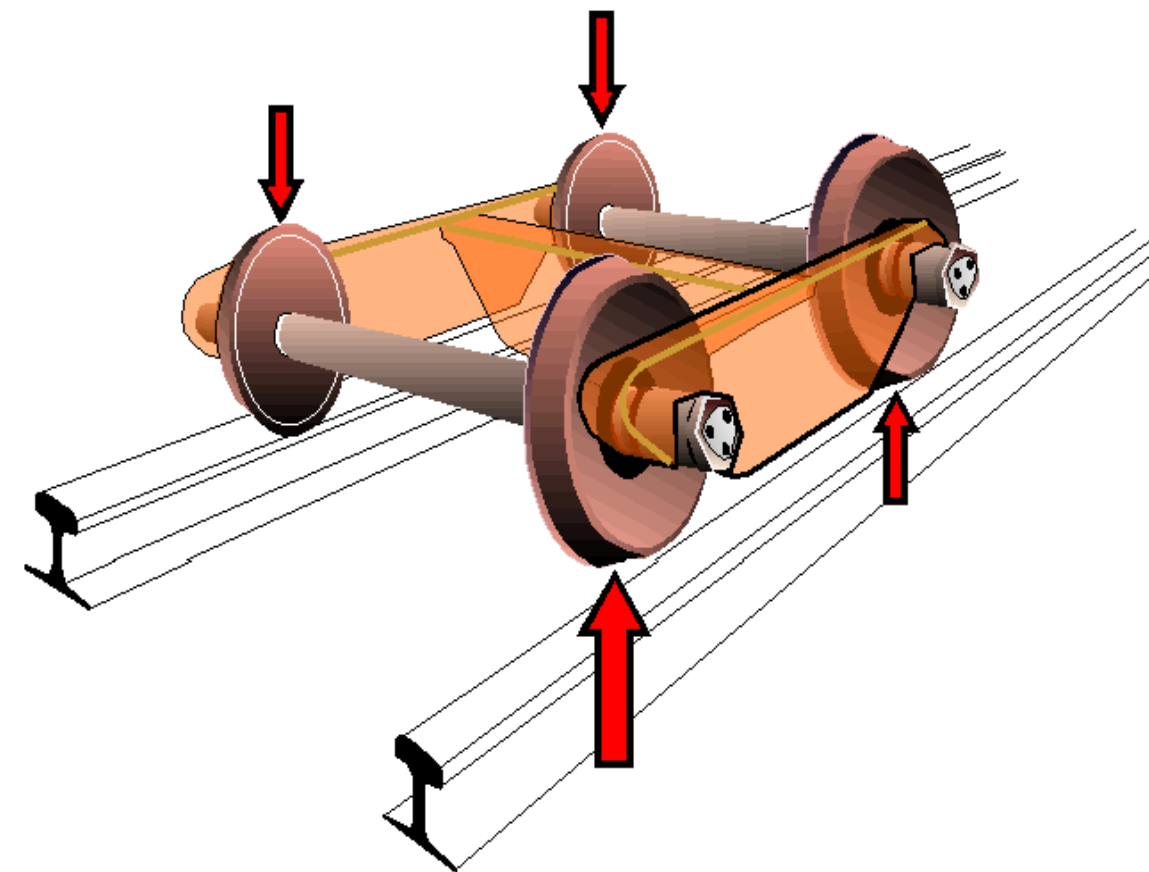
Low Rail

Load Equalization & Truck Design

3-piece Truck

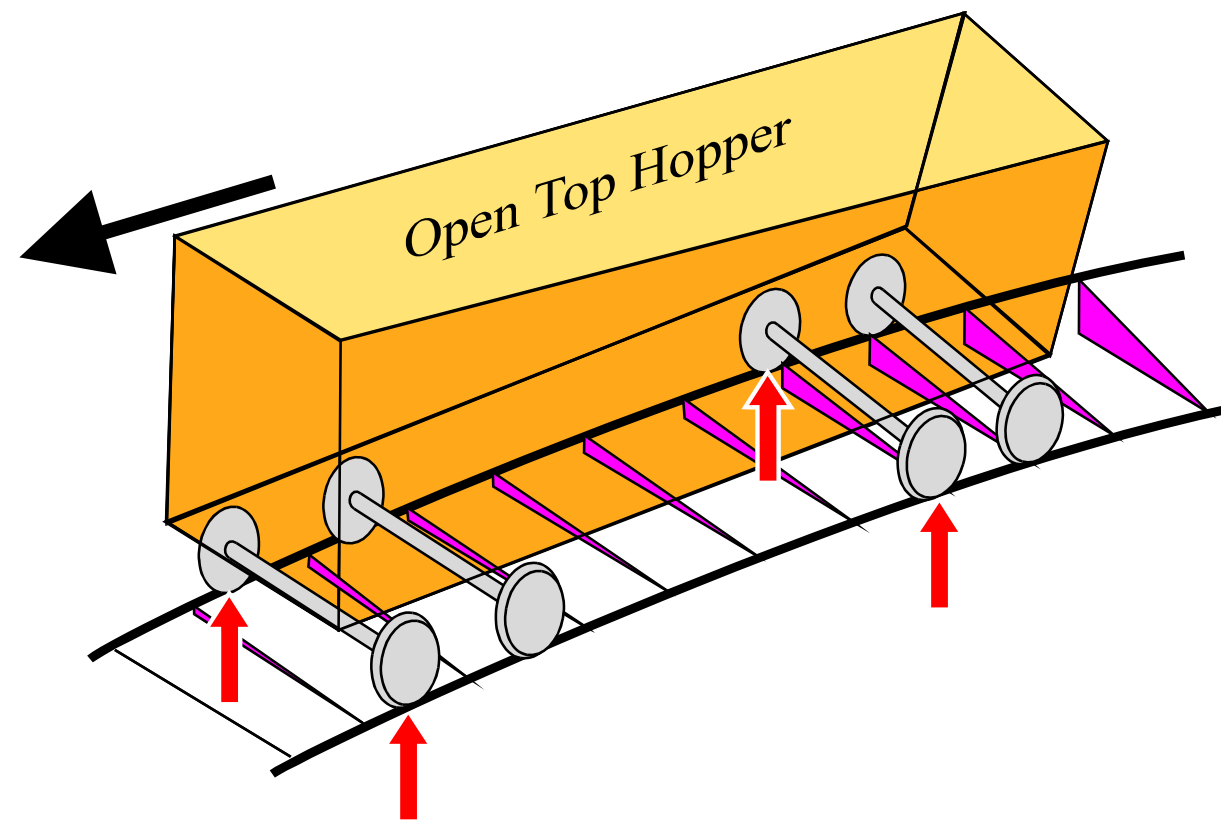


H-frame Truck with
Stiff Primary Suspension

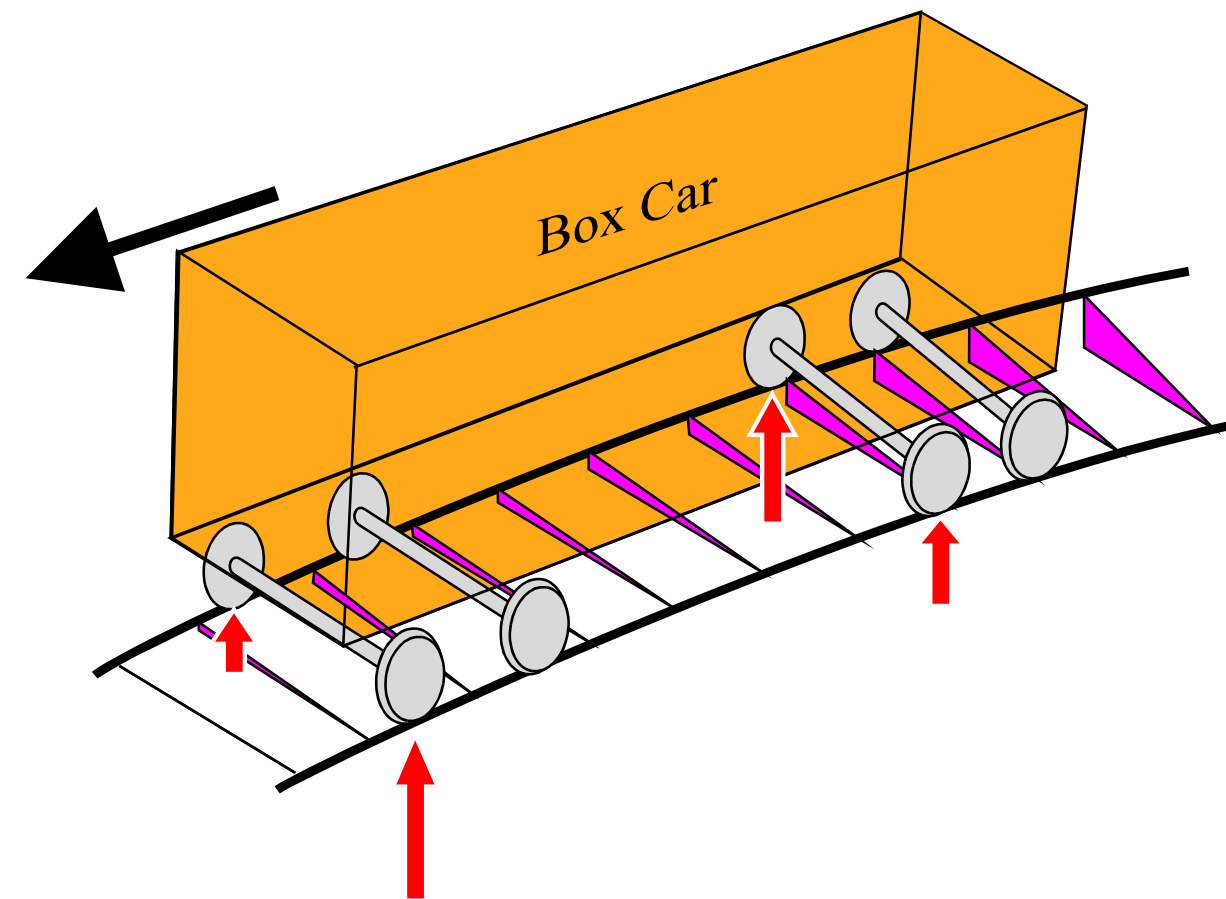
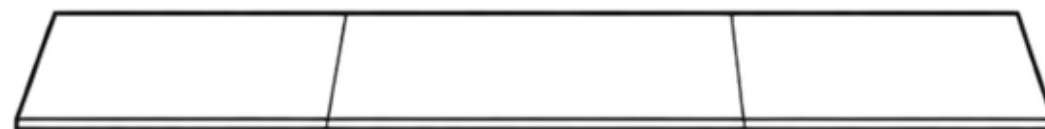


Carbody Flexibility / Torsional Stiffness

Track Twist / Track Warp & Vertical Wheel Loads



Flexible Carbody
Examples: Hopper, Flatcar
Like an Unrolled Paper Towel Tube

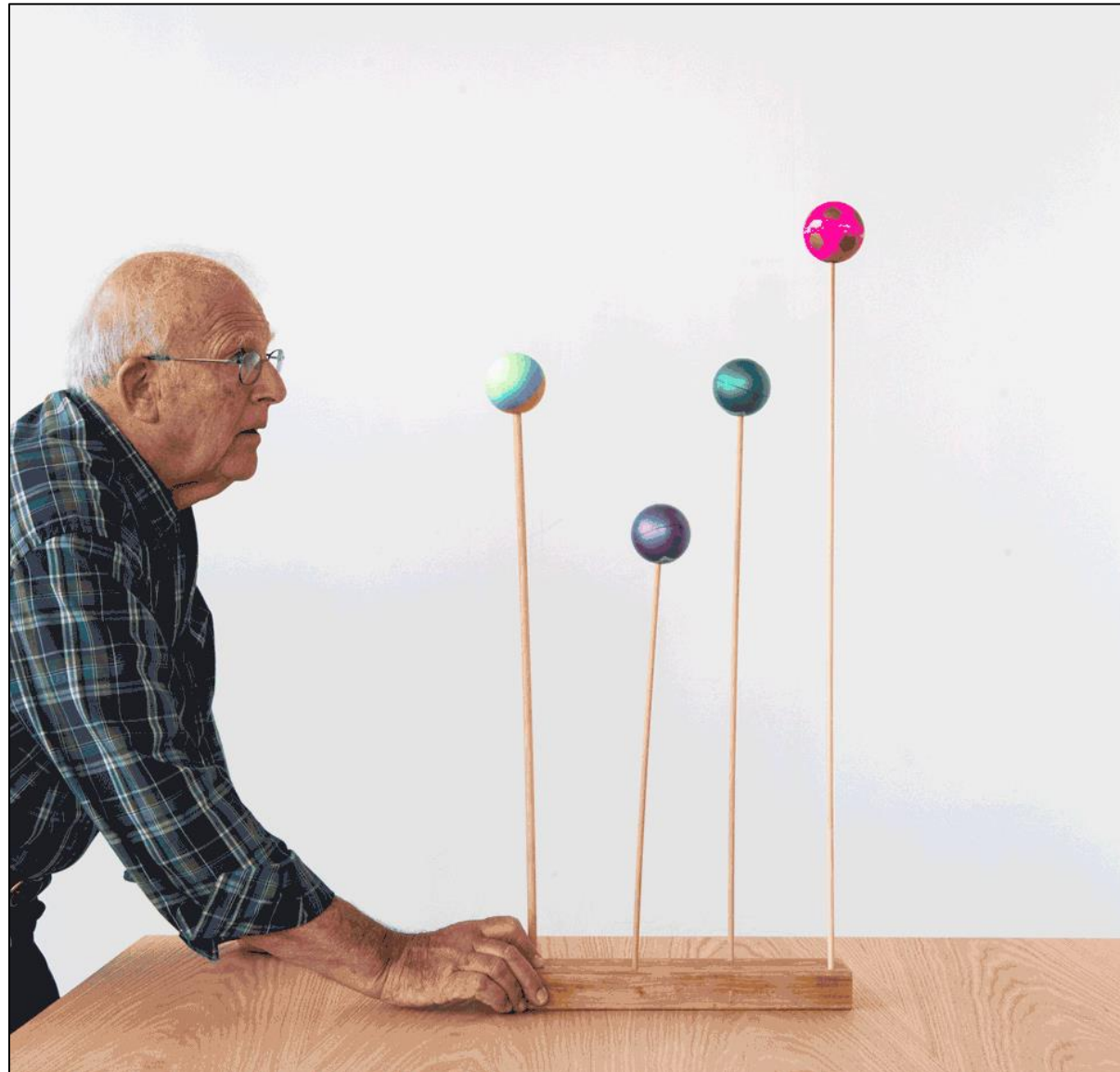


Stiff Carbody
Examples: Box Car, Tank Car
"Paper Towel Tube"





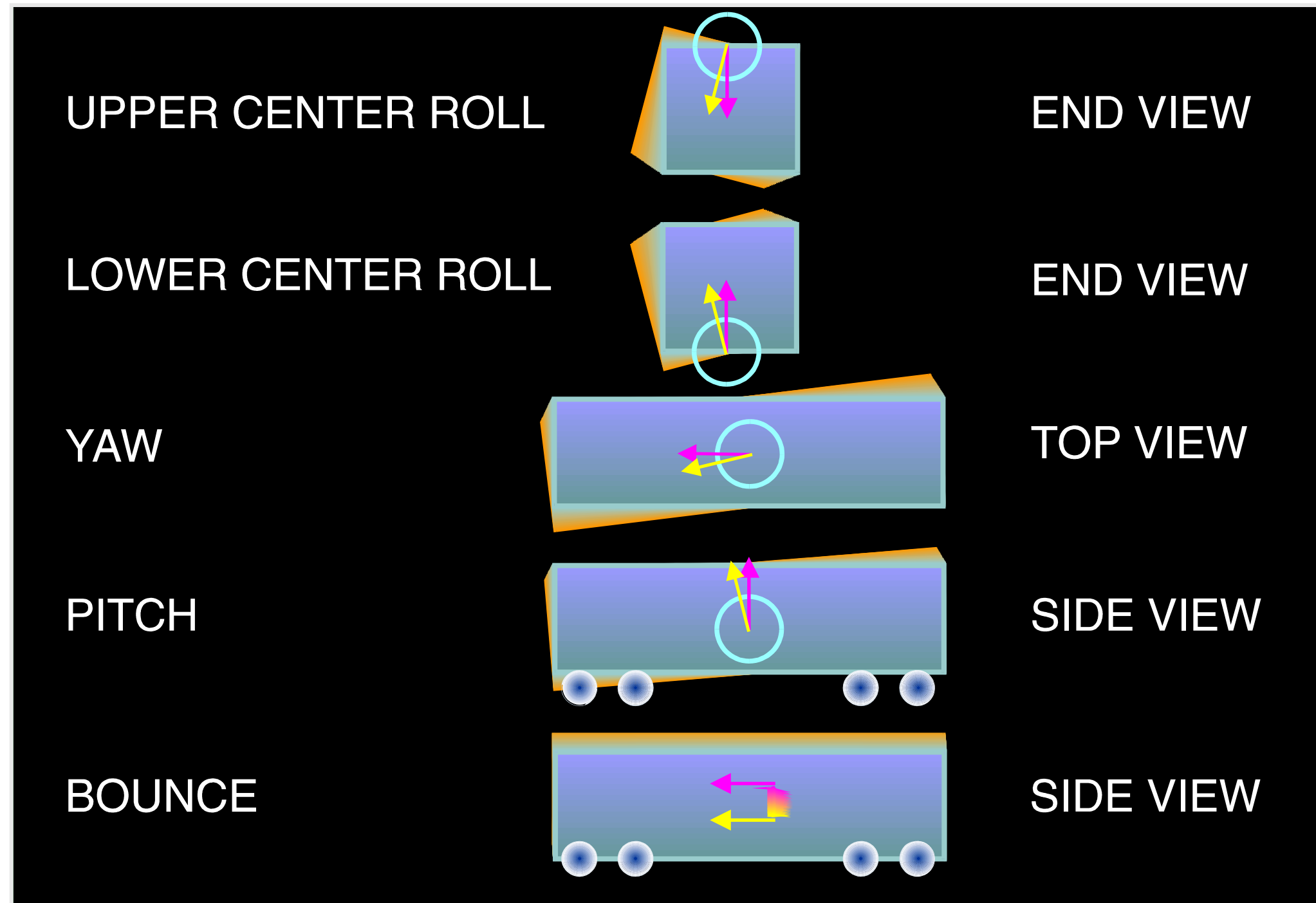
(4) Resonance in the Vehicle/Track System Designing and Maintaining for the ‘Sweet Spot’



<https://www.youtube.com/watch?v=gR3NSyKHsEg>



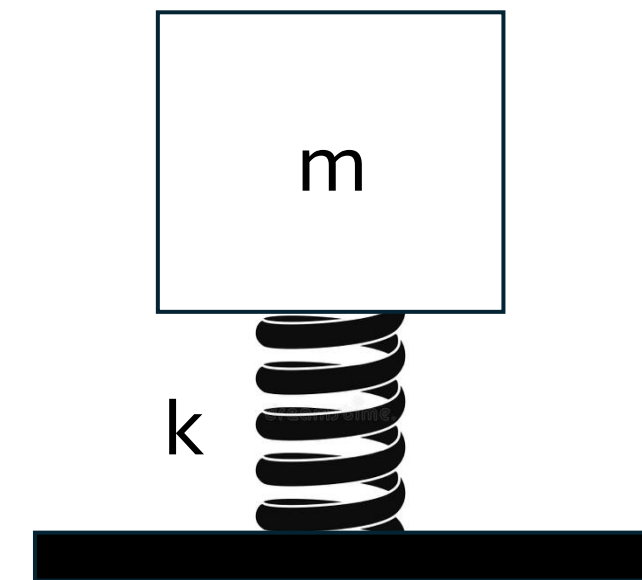
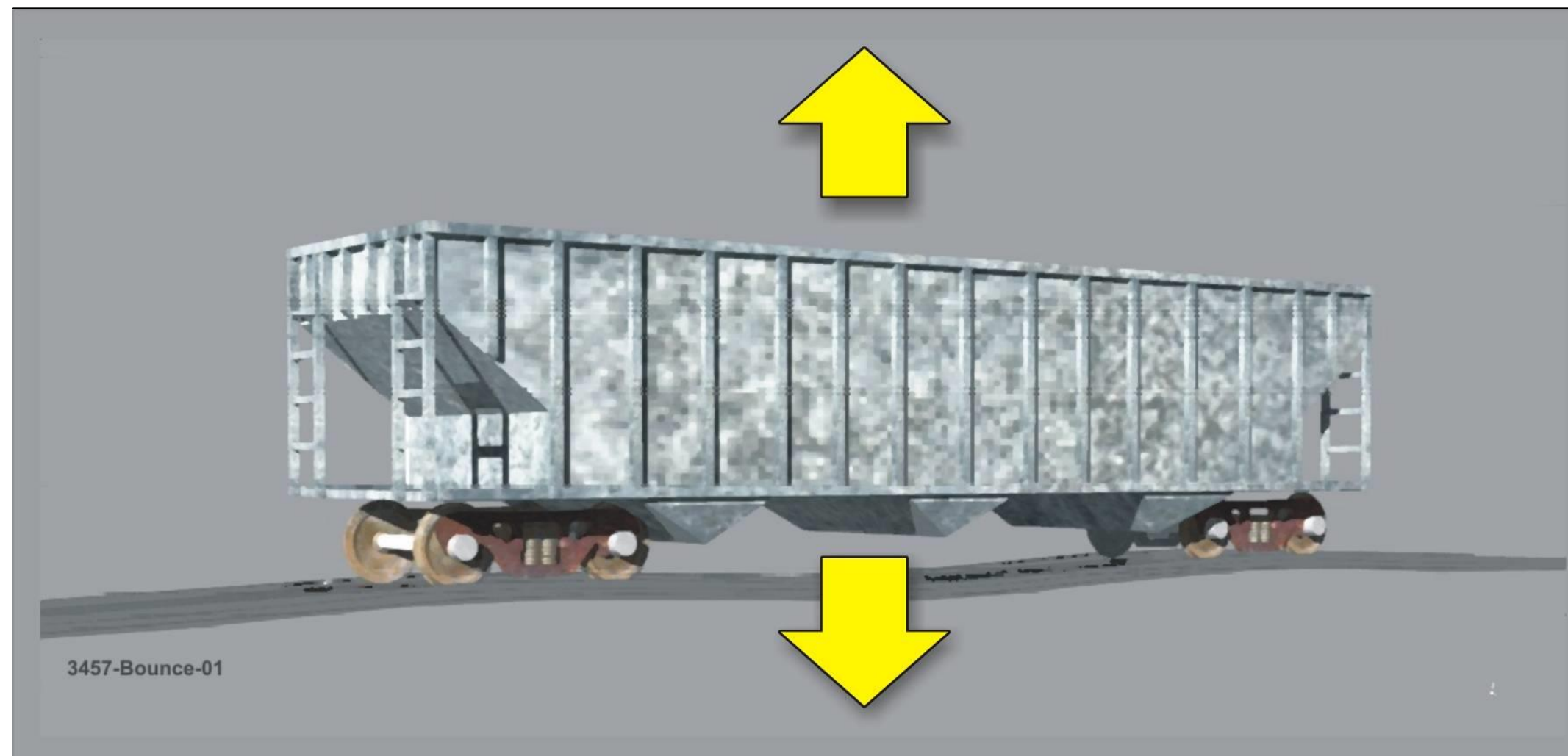
Rigid Body Modes





Bounce Mode

The bounce mode is excited when the wavelength of several parallel dips in the track match the truck center distance, and the speed aligns with the critical frequency of the vehicle system.

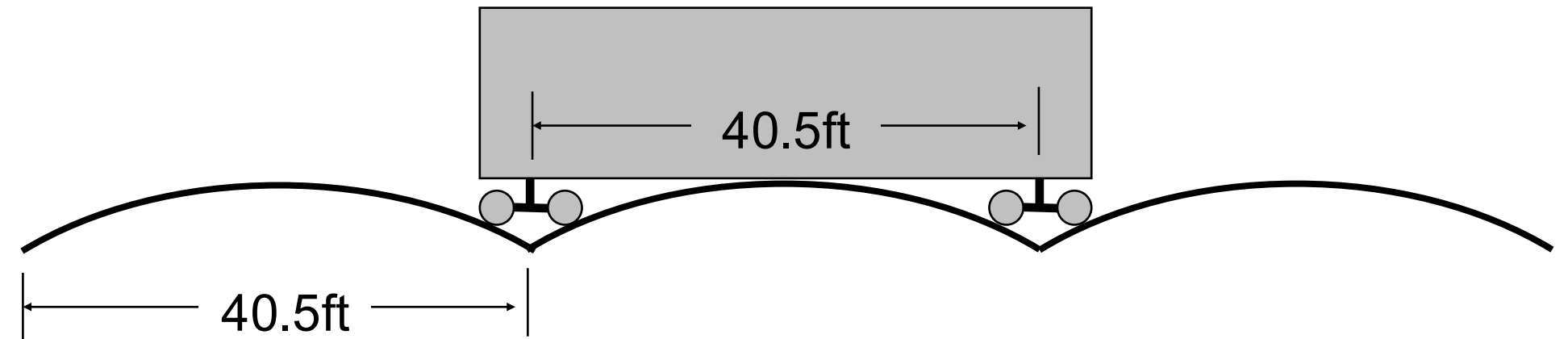


m : Mass of object
 k : Spring stiffness



Bounce Mode

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$



If $f = 1.96 \text{ Hz}$

What is the critical bounce speed?

$40.5 \text{ ft} * 1.96 \text{ Cycles / Sec} \approx 80 \text{ ft/s}$

$80 \text{ ft/s} = 55 \text{ mph}$



If the goal is to operate at higher speed with less risk of exciting the bounce mode...

f increases with:

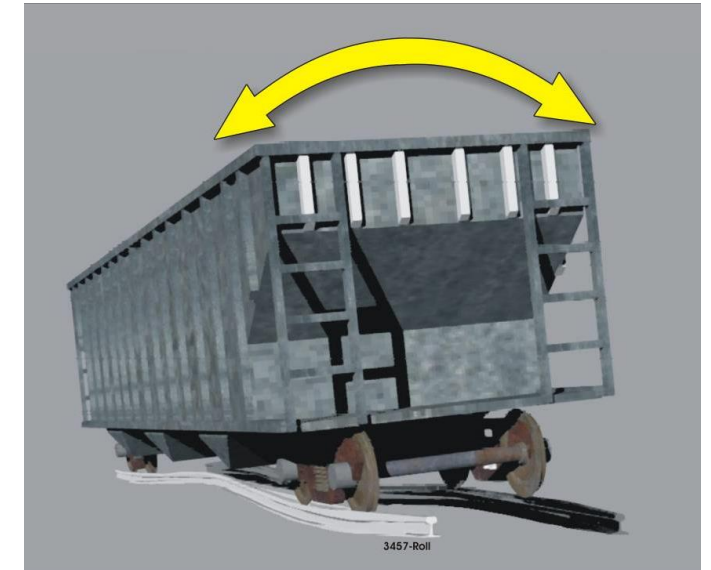
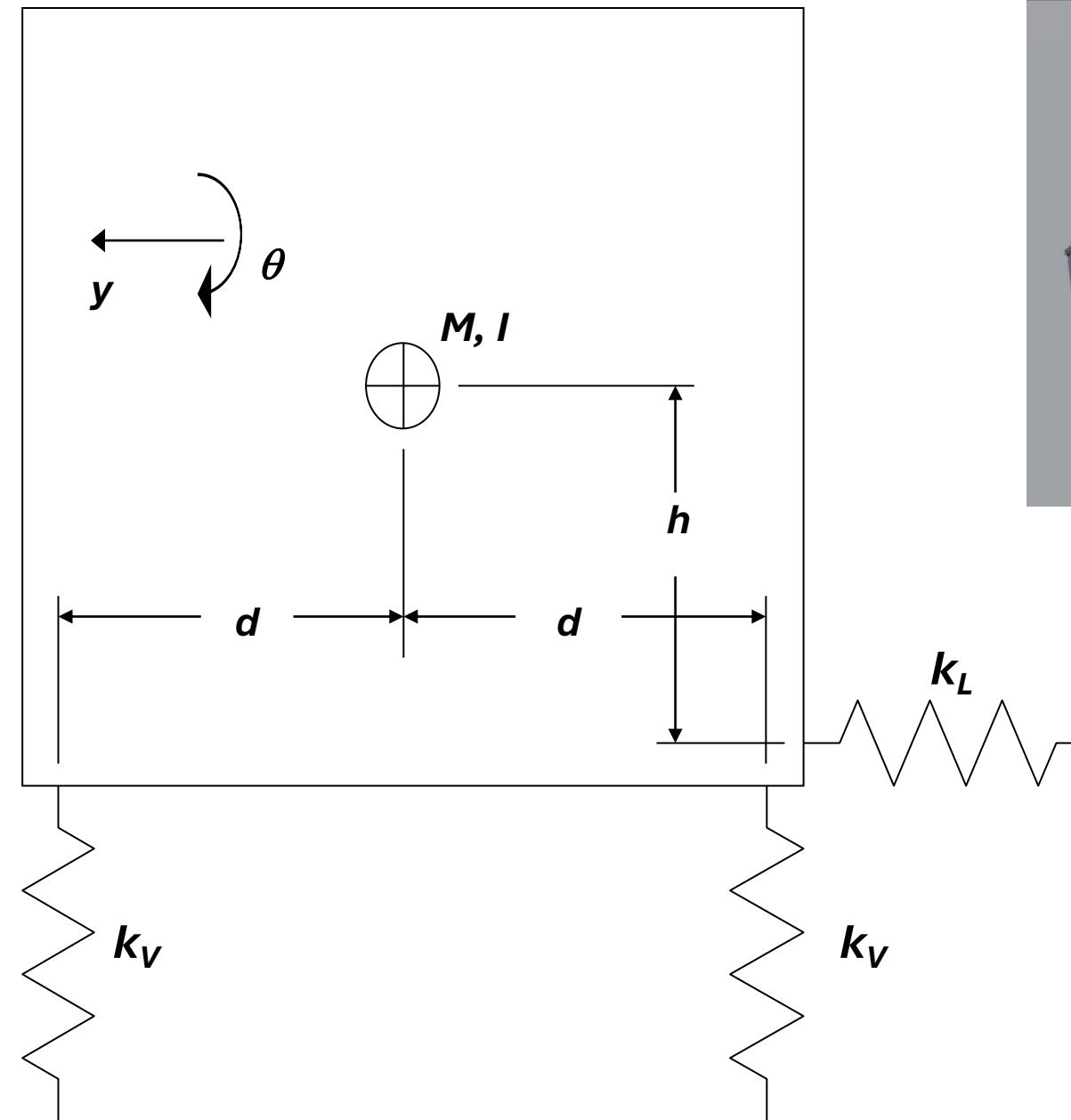
↑ k Stiffer vertical suspension

↓ m Lighter weight



Roll Mode

- Coordinates
 - y lateral
 - θ roll
- k_V - Vertical Stiffness
- k_L - Lateral Stiffness
- M - mass
- I - roll inertia
- Ignoring vertical DOF but k_V 's make a roll stiffness as shown previously of $k_{roll} = 2k_V d^2$





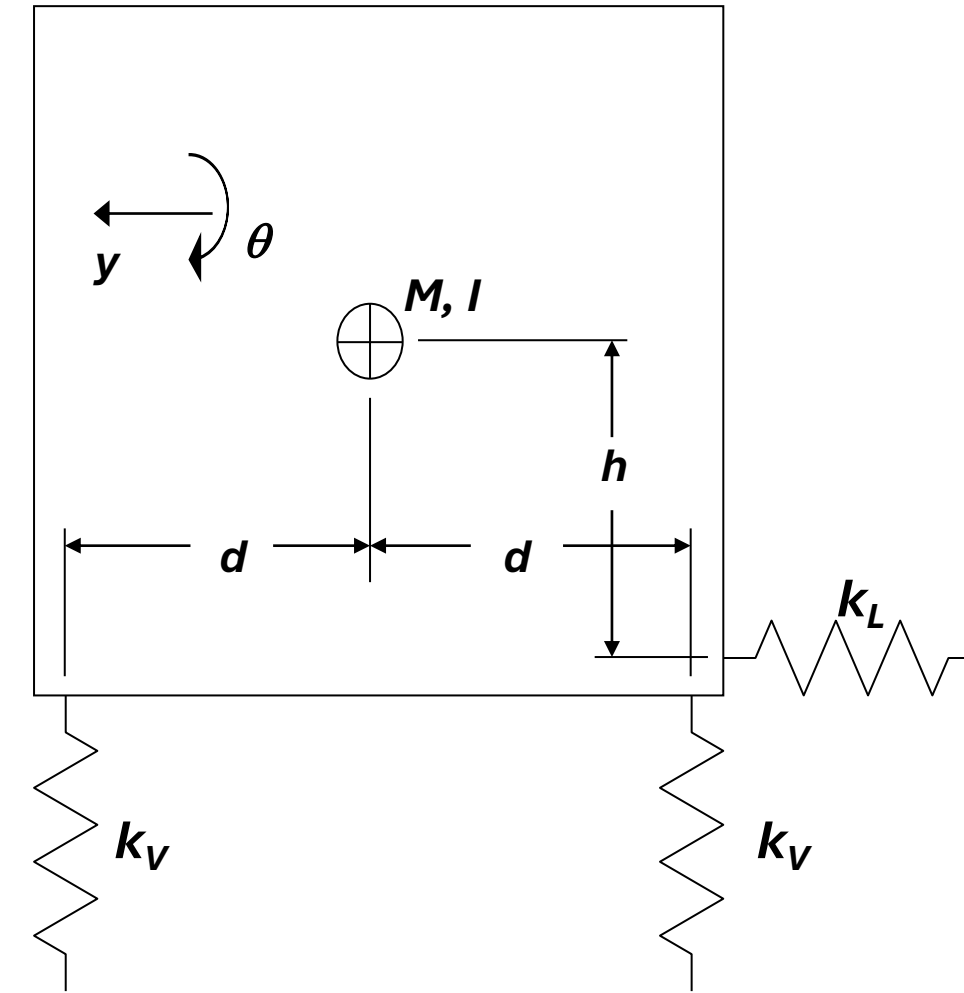
Solving for Critical Roll Frequency More Complex

Differential Equations:

$$\sum \text{Lateral Forces} = -k_L y - k_L \theta h = M \ddot{y}$$

$$\sum \text{Moments about CG} = -k_L y h - (k_L h^2 + k_{roll}) \theta = I \ddot{\theta}$$

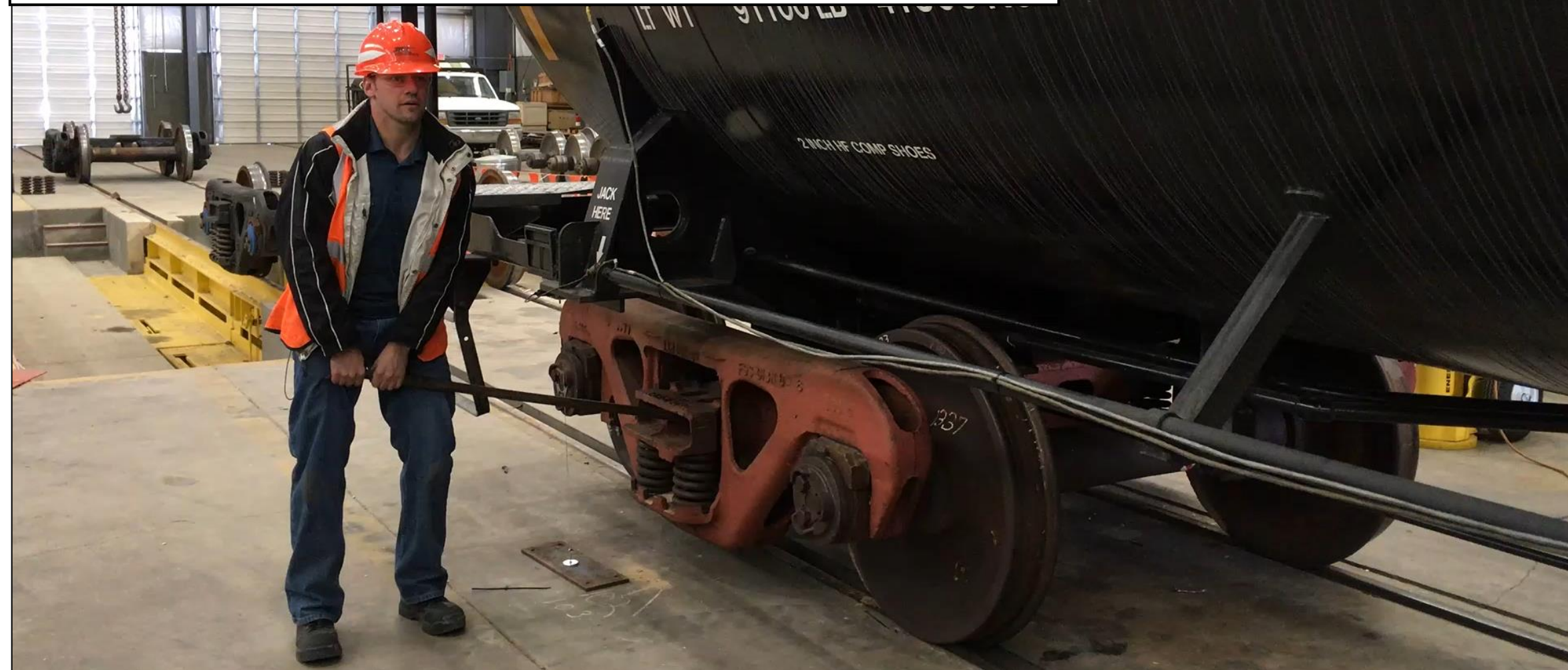
Proceed with an eigenvector analysis to solve for natural frequencies, or...





Test & Measurement Vehicle Characterization or “Shake & Bake”

1) Exciting the Roll Mode

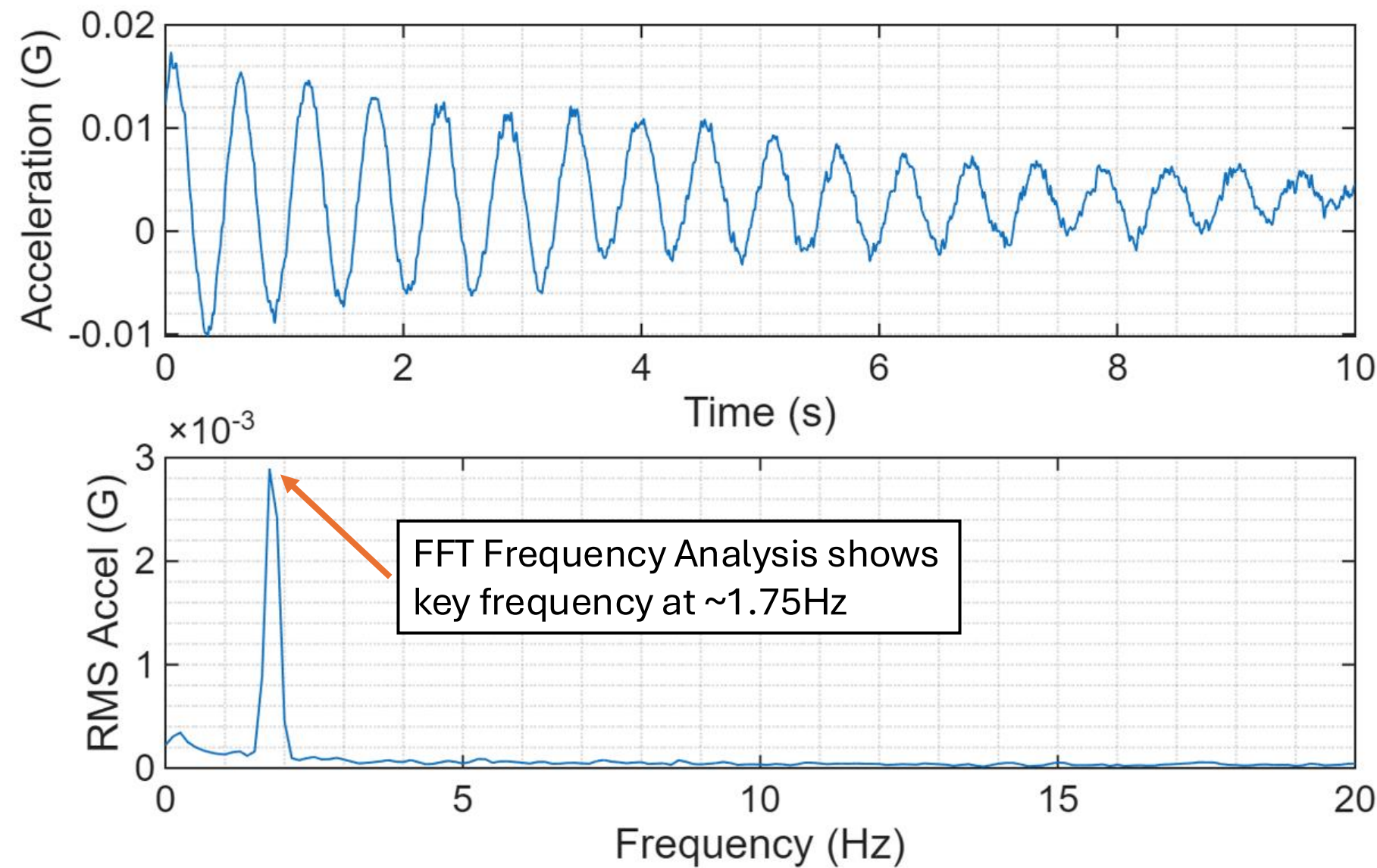


2) Yaw Mode



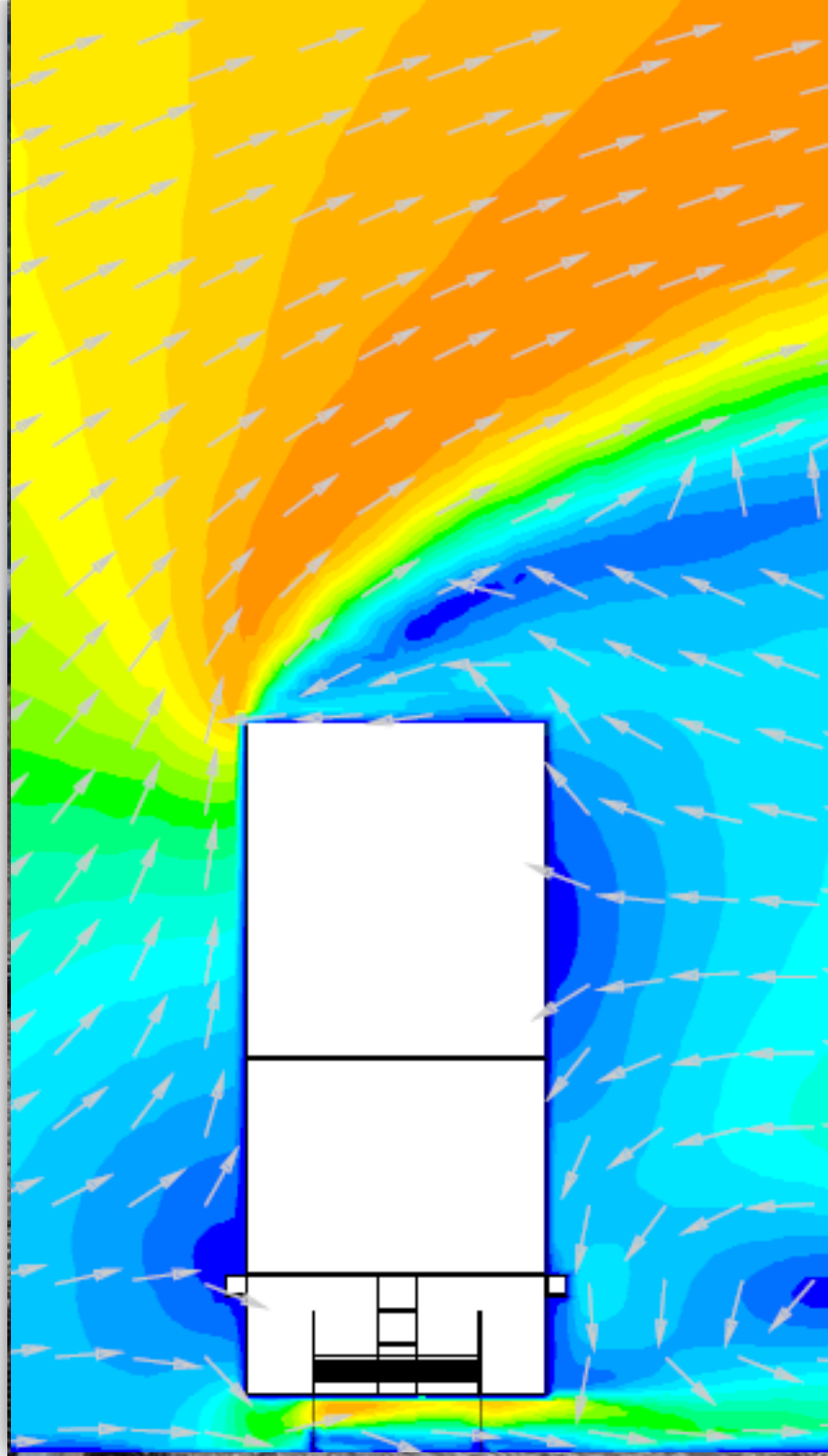


Sample Characterization Data (Yaw Mode) Acceleration & Frequency Analysis



In general:

1. Loaded cars have lower resonant frequencies
 - a) ~1-5 Hz
 - b) Empty cars: 3-8 Hz
 - c) Lowest – lower center roll
 - d) Highest – upper center roll

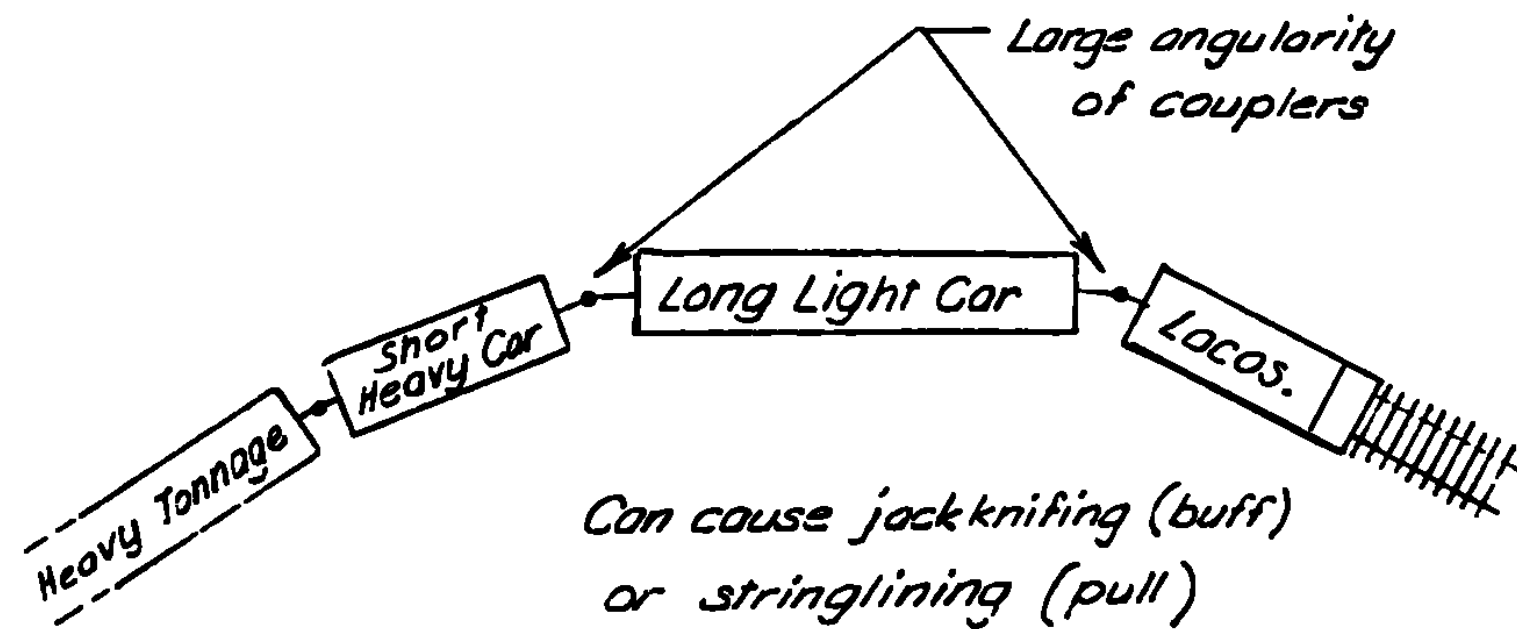


(5) External Forces



Train Dynamics Overview

Coupler Forces



1. Key sources of coupler forces
 - Grade
 - Train handling (Tractive / braking)
 - In-train velocity differences (slack)
2. Coupler forces with a lateral component affect wheel/rail forces
 - Track curvature
 - Car geometry (long/short)
3. Lighter cars have less vertical wheel load



Environmental Forces

High Winds



BNSF wind fence at Abo Canyon, Courtesy Railway Track & Structures

- Light cars with large surface area have greater risk of blowover in high crosswinds
- Train speed can affect wind load depending on wind direction
- Wind fences and forecasting as mitigation



Summary

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2. High Speed Stability
3. Curving & Load Equalization
4. Resonance in the Vehicle-Track System
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